Supervised by:
Dr. Mohammed Sharkawy

Obstetrics

2026

Out of the League
Family Medicine

Prepared by: Dr. Fatimah Alfaddagh

3rd Edition

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TOPIC: Pre-conception advice and management

Definition

- Pre-conception care is defined by the World Health Organization as providing biomedical, behavioural, and social health interventions before pregnancy occurs to improve health and modify behaviours and environmental influences. The goal is to improve the long and short-term health outcomes of women and their children.
- It should be considered as a continuum rather than a single visit.
- Preconception care is important because:
    - Women's health at conception and in early pregnancy impacts on their lifelong health and that of their children.
    - It allows physical and mental health conditions and social needs to be addressed and managed prior to pregnancy.
    - It allows women to be aware of potential risks and make an informed decision about their pregnancy.
    - Many potentially modifiable risk factors which influence pregnancy outcomes are present before conception. This means that prenatal care is often given too late to change the outcome of the pregnancy.

Assessment

From age 16 years to 45 years (Female).

What do I need to assess in a woman planning a pregnancy?

Consider a discussion about pre-conception issues of women of childbearing age when appropriate during a consultation.

Each part of the assessment identifies a specific management requirement. Further information about how to assess and manage individual components is provided by following the hyperlinks.

- **Assess the following** in a woman who is planning to become pregnant:
    - Plans for timing of pregnancy.
    - Previous obstetric history.
    - Dietary habits, height, weight, and body mass index (BMI).
    - Understanding of when to take folic acid and what dose is needed (as determined by the risk of a neural tube defect).
    - Cervical smear status.
    - Smoking status.
    - Amount of alcohol consumed.
    - Use of recreational drugs.
    - Risk of hepatitis B or HIV infection.
    - Immunity to rubella and chickenpox.
    - Concerns about work exposure to hazardous substances and radiation.
    - Use of prescription medications, over-the counter medicines, or herbal remedies.
    - Plans to travel to an area with active Zika virus transmission.
    - Chronic health problems. Pre-conception management advice is provided for:
        - Mental health issues
        - Epilepsy
        - Diabetes mellitus
        - Thyroid disease
        - Chronic hypertension
        - Chronic cardiac disease
        - Renal disease

        - Asthma
        - Previous thromboembolism
        - Rheumatological conditions
        - Inflammatory bowel disease
        - Haemoglobinopathies
    - Risk of having a baby with an inherited genetic disorder.

Scenario: Pre-conception advice for all women
From age 16 years to 45 years (Female).
What advice can I give a woman about the timing of pregnancy?

- **Discuss the potential impact of maternal age on fertility and birth outcomes.**
    - Women over 35 years have an increased risk of miscarriage, chromosomal abnormalities, and obstetric complications compared to younger women.
- **Discuss interpregnancy interval** (the time from delivery of one child to conception of the next).
    - There is some evidence to suggest that an interpregnancy interval of 18–59 months is safer in terms of perinatal outcomes, but the decision should take into account the woman's individual circumstances (for example a shorter interpregnancy interval may be appropriate for older women concerned about age-related decline in fertility).
- **If the woman has had a previous miscarriage, discuss that there is no definite 'right time' to start trying to conceive again.** The decision will be influenced by a number of factors including when the woman and her partner feel ready; speed of physical recovery; and whether the woman is awaiting test results or being followed up after surgery, or ectopic or molar pregnancy.

What advice can I give a woman about conception?

- Advise that:
    - Of 100 couples (in which the woman is aged under 40 years) having regular sexual intercourse without contraception:
        - More than 80 will conceive within 1 year.
        - About half of those who do not conceive in the first year will do so in the second year.
    - The remainder will take longer and some of these may need help for them to conceive.
- Advise that sexual intercourse every 2 to 3 days optimises the chances of pregnancy.
    - There is no need to plan intercourse to coincide exactly with ovulation — this does not increase the chances of success and can cause stress for the couple.
- Advise women planning pregnancy who have been using the progestogen-only injection for contraception that normal fertility may be delayed for up to 1 year after the last injection.

What advice should I give a woman planning pregnancy regarding folic acid?

- **Assess the couple's risk of a neural tube defect (NTD)**.
    - Couples are at high risk of conceiving a child with an NTD if:
        - Either partner has an NTD, they have had a previous pregnancy affected by an NTD, or they have a family history of an NTD.
        - The woman is taking anti–epileptic medication.
        - The woman has diabetes mellitus**,** sickle cell anaemia, or thalassaemia.
        - The woman is obese (defined as a body mass index [BMI] of 30 kg/m^2or more).

- **Advise women who are at normal risk for an NTD** to take of folic acid 400 micrograms daily, and once pregnant, to continue this until the twelfth week of pregnancy.
- **Advise women at high risk of an NTD** to take folic acid 5 mg daily and, once pregnant, to continue this until the twelfth week of pregnancy.
    - Women with sickle cell disease, thalassaemia, or thalassaemia trait should take folic acid 5 mg daily throughout pregnancy.

What advice should I give on diet?

- Advise women considering pregnancy to eat a healthy, balanced diet.
- To maintain a healthy weight before pregnancy, women should:
    - Base-meals on starchy food (for example bread, rice, pasta, potatoes), choosing wholegrain if possible.
    - Eat fibre-rich foods (for example fruit, vegetables, oats, beans, peas, lentils).
    - Eat at least 5 portions of different fruits and vegetables each day.
    - Eat a low-fat diet.
    - Eat as little as possible of fried food, drinks and confectionary with added sugar (for example cakes, fizzy drinks), and other foods high in fat and sugar.
    - Eat breakfast.
    - Be aware of portion sizes of meals and snacks, and how often they eat.

What should I advise on weight management?

- Advise women that achieving a healthy weight (BMI 18.5–24.9 kg/m^2) before becoming pregnant reduces the risk of pregnancy complications.
- Advise the woman of the potential health risks of being obese (BMI of 30 kg/m^2 or more) including:
    - Reduced fertility.
    - Increased risk of miscarriage.
    - Gestational diabetes.
    - Gestational hypertension/pre-eclampsia.
    - Macrosomia and shoulder dystocia.
    - Preterm delivery.
    - Birth trauma.
    - Caesarean delivery.
    - Postpartum complications (for example haemorrhage, thrombosis and infection).
    - Stillbirth.
    - Congenital anomalies (for example neural tube defects, cardiovascular anomalies, cleft palate, limb reduction, anorectal atresia, hydrocephaly).
- Advise and encourage women who are obese (BMI of 30 kg/m^2 or more) to lose weight before becoming pregnant.
    - Women should be informed that losing 5–10% of their weight (a realistic target) would have significant health benefits, and could increase their chances of becoming pregnant.
    - Women should be encouraged to check their weight and waist measurement periodically, or as an alternative, check the fit of their clothes.
    - Offer a weight loss support programme that includes advice about diet and physical activity.
    - Women should be aware that if they do become pregnant, there is no need to 'eat for two' or to drink full-fat milk.
- Advise women who are obese (BMI of 30 kg/m^2 or more) to take folic acid 5 mg daily starting at least one month before conception, and continuing during the first trimester.

- Advise women with a low BMI (less than 18.5 kg/m^2) of the potential health risks of being underweight, including:
    - Reduced fertility.
    - First-trimester miscarriage.
    - Preterm birth.
    - Low birth weight.
    - Gastroschisis.

What advice should I give to a woman planning pregnancy who smokes?

- **Advise all women planning pregnancy who smoke to stop smoking.**
    - Offer women who wish to stop smoking referral to a smoking-cessation service.
    - Advise women who may become pregnant to initially try to stop smoking without using nicotine replacement therapy (NRT).
    - Consider offering NRT to women who are planning a pregnancy, and who have tried and failed to stop smoking without using NRT.
- Do not prescribe bupropion or varenicline to women who may become pregnant.

What advice should I give to a woman planning pregnancy, regarding alcohol consumption?

- Advise women planning pregnancy (or who are at any stage of pregnancy) to avoid drinking alcohol.
- Offer specialist referral if a woman is unable to reduce her drinking with support in primary care.

What advice should I give to a woman planning pregnancy, who uses illicit drugs?

- Advise women planning pregnancy who use illicit or recreational drugs (including so-called 'legal highs'), to stop using them if they are able to do so.
- Offer women planning pregnancy who use illicit drugs and are unable to stop with support in primary care referral to a specialist service.
- Offer contraceptive advice to women using illicit drugs who may become pregnant before illicit drug use has stopped.
- Offer women injecting illicit drugs testing for hepatitis B, hepatitis C, and HIV.

What advice should I give to women who are planning pregnancy who may be exposed to hazardous substances or radiation?

- Advise women planning pregnancy to be aware of the potential for exposure to toxic substances in their home, workplace, and surrounding environment and to avoid them if possible.
    - It is possible for chemical exposure to occur through breathing, eating or drinking, or skin absorption.
- Advise a woman who is planning pregnancy and is concerned about work exposure to hazardous substances, infections, or radiation, to discuss her intention of becoming pregnant to her employer, if possible.
- Advise a woman planning pregnancy who does not wish to discuss her intention to become pregnant to her employer, that information about the risk of exposure to specific substances can be obtained by contacting an expert at the Health and Safety Executive.
- It may be possible at some work places to have a discussion with an occupational medicine specialist, if this is available.

What advice should I give about prescription and over-the-counter medication and herbal remedies?

- If the woman is taking prescribed medication, discuss any changes that may need to be made, taking into account:
    - It is important to continue certain drugs (for example if stopping the drug would cause a worsening of the underlying disease that would be considered to be a higher risk to the pregnancy).
    - Some medications are not considered to be safe in pregnancy and may potentially adversely affect the fetus.
    - There may be a need to switch to a safer alternative medication before conception.
    - Some medications will need to be stopped if they require a washout period before conception.
    - Ideally, the smallest number of medications at the lowest dose possible should be used when trying to conceive.
- Advise women planning pregnancy not to take any over-the-counter medicines without consulting a pharmacist to ensure that these products are safe to take if she were to become pregnant.
- Advise women planning pregnancy not to take any herbal remedies.

What advice should I give a woman who wishes to become pregnant regarding cervical screening?

- Advise all women planning pregnancy who are due a cervical smear test to have the test as soon as possible, *before* becoming pregnant.

What advice should I give women planning pregnancy regarding immunizations?

- **Determine if a woman planning pregnancy is protected against rubella** (for example, documentation of having received two doses of rubella-containing vaccine, or a positive antibody test for rubella). Offer measles, mumps, and rubella (MMR) vaccine to seronegative women planning a pregnancy.
    - Note that MMR vaccine should not be given to immunocompromised or pregnant women, and women who are not pregnant should avoid pregnancy until one month after administration of the vaccine.
- **Determine if a woman planning pregnancy has immunity to varicella** (if there is a definite history of chickenpox or herpes zoster, she can be considered to be protected).
    - If there is not a definite history of chickenpox or shingles, and the woman is eligible for the vaccine (for example healthcare workers who come into direct contact with patients; laboratory staff where exposure to varicella virus is an occupational risk; and healthy, susceptible close household contacts of immunocompromised patients), offer serological testing and, if found to be without varicella zoster antibody, offer vaccination.
        - Note that the varicella vaccine should not be given to immunocompromised or pregnant women, and women who are not pregnant should avoid pregnancy until one month after administration of the last dose.
- **Vaccinate women planning pregnancy against hepatitis B if they are at high risk of contracting the disease.**
    - People at risk include intravenous drug users, those who change sexual partners frequently, those with chronic renal or liver disease, and those who are in close contact with people with hepatitis B.

What pre-conception advice should I give women about Zika virus infection?

- Advise the woman to avoid becoming pregnant during a trip to an area with active Zika virus transmission.
- Advise on mosquito bite avoidance measures, day and night (particularly mid-morning and late afternoon to dusk).
- Advise that on returning to the UK:
    - If her male partner did not travel, she should avoid conception and consider the use of barrier precautions for 2 months from symptom onset or departing a Zika-affected country.
    - If her male partner did travel, she should avoid conception and consider the use of barrier precautions for 3 months from symptom onset or departing a Zika-affected country.

Women with mental health problems
From age 16 years to 45 years (Female).
How should I manage a woman with a mental health problem who wants to become pregnant?

- **Discuss with the woman:**
    - How pregnancy and childbirth could affect her mental health problem (for example risk of relapse).
    - How her mental health problem and/or its treatment might affect her or her baby, before and after birth.
    - The risks of not treating her condition and the importance of controlling symptoms before conception.
- **Consider referring women with a current or past severe mental health problem** (for example severe depressive disorders, bipolar disorder, and psychosis) to secondary care for pre-conception counselling. Ideally refer to a specialist perinatal mental health service, if available.
    - Advise the woman to continue using effective contraception until a full assessment by the psychiatrist has taken place.
    - Advise the woman not to stop taking her medication unless otherwise directed by the psychiatrist.
- **Manage women who do not require referral,** using clinical judgement and taking into account the woman's past history, and her preferences.
    - Treatment options include continuing the woman's current drug regime, switching to a regime with a lower risk of adverse effects, or stopping treatment (avoid abrupt discontinuation of medication).
        - Advise the woman not to stop taking her medication unless otherwise directed by a clinical specialist in mental health.
    - Discuss the potential benefits of psychological interventions and psychotropic medication, the consequences of altering or stopping treatment, and potential harms of treatment.
    - When prescribing psychotropic medication for women planning a pregnancy, consider the latest data on risks to the fetus and baby.
    - Seek specialist advice if in doubt about the risks of the mental health problem or the benefits and harms of treatment in pregnancy, preferably from a specialist perinatal mental health service.
- Measure prolactin levels in women planning a pregnancy who are taking prolactin-raising antipsychotic medication.

Women with chronic medical conditions
From age 16 years to 45 years (Female).
How should I manage a woman with diabetes mellitus who wishes to become pregnant?

- **Refer all women with diabetes mellitus who wish to become pregnant to a pre-conception diabetes clinic (if available) or to their diabetes care team,** as soon as possible.
    - Advise women planning a pregnancy that their blood glucose targets and current medication for diabetes and its complications will need to be reviewed.
- **Discuss the effect of diabetes on pregnancy, and pregnancy on diabetes.**
    - Pregnancy is associated with increased rates of hypoglycaemia (and decreased hypoglycaemic awareness), increased rates of diabetic ketoacidosis, and worsening of diabetic retinopathy and nephropathy.
    - Diabetes during pregnancy is associated with an increased risk of a number of adverse outcomes, including miscarriage, macrosomia and obstetric complications, developmental and growth abnormalities, and stillbirth.
- **Discuss the importance of optimal blood glucose control to reduce the risk of miscarriage, congenital malformation, stillbirth, and neonatal death.**
    - Women planning pregnancy should ideally aim to achieve a pre-conception HbA1c value of less than 48 mmol/mol (less than 6.5%) if this can be achieved safely. Any reduction in HbA1c towards the normal HbA1c value of less than 48 mmol/mol (less than 6.5%) is likely to reduce the risk of congenital malformations.
    - Capillary plasma glucose target ranges should be considered to be the same as those recommended for all people with type 1 diabetes (a fasting plasma glucose level of 5–7 mmol/L on waking and a plasma glucose level of 4–7 mmol/L before meals at other times of the day). Goals for self-monitored glucose should take into account the risk of hypoglycaemia**.**
    - Effective contraception methods should be used until a woman's individualized target has been achieved.
- **Ensure that the woman is offered:**
    - Individualized advice on diet and, if her body mass index is more than 27 kg/m^2, advice on weight loss.
    - Advice to take folic acid 5 mg daily until week 12 of pregnancy.
    - Retinal assessment (unless she has had an annual retinal assessment in the last 6 months). Rapid optimisation of blood glucose levels should be avoided until assessment and any necessary treatment has been undertaken.
    - Renal assessment, including measuring for albuminuria, before stopping contraception. Referral to a nephrologist should be considered if serum creatinine is 120 micromol/L or more; the urinary albumin:creatinine ratio is more than 30 mg/mmol; or the estimated glomerular filtration rate (eGFR) is less than 45 mL/minute/1.73m^2.
- Measure thyroid stimulating hormone (TSH), free thyroxine, and thyroid peroxidase antibodies in women with type 1 diabetes.
- Discuss other standard pre-conception measures.

How do I manage a woman with epilepsy who wishes to become pregnant?

- **Refer all women with epilepsy to a specialist before conceiving.**
    - Advise the woman to continue using effective contraception until a full assessment by the specialist has taken place.
    - Advise the woman not to stop taking her medication unless otherwise directed by the specialist.
- Discuss the standard pre-conception measures.

How do I manage a woman with chronic cardiac disease who wishes to become pregnant?

- **Refer all women with cardiac disease who wish to become pregnant to a cardiologist** for accurate diagnosis and functional assessment so that maternal and fetal risk can be assessed.
    - Advise the woman to continue using effective contraception until a full assessment by the cardiologist has taken place.
    - Advise the woman not to stop taking her medication unless otherwise directed by the cardiologist.
- Discuss the standard pre-conception measures.

How do I manage a woman with chronic hypertension who wishes to become pregnant?

- **Refer all women with a history of hypertension who wish to become pregnant** to a specialist in hypertensive disorders prior to becoming pregnant.
    - Advise the woman that she will be at increased risk of pre-eclampsia during pregnancy and that she will require careful monitoring.
    - Advise the woman to continue using effective contraception until she has been fully assessed.
    - Advise the woman not to stop taking her current medication unless directed by the specialist.
- Discuss the standard pre-conception measures.

How do I manage a woman with asthma who wishes to become pregnant?

- **For women with mild asthma,** ensure that asthma is well controlled.
- **For women with moderate or severe asthma and those in whom asthma is poorly controlled,** refer to a chest physician to ensure adequate control and monitoring.
- **Discuss the importance of continuing to take asthma medication as prescribed,** both before conception and throughout the pregnancy, to maintain good asthma control.
    - Treatment of controlled asthma requires little modification in pregnancy. The risks from uncontrolled asthma are much greater than the risk from asthma treatment during pregnancy.
    - Steroid tablets should be used as normal in the pre-conception period and during pregnancy and never withheld because of pregnancy.
- Discuss the standard pre-conception measures.

How should I manage a woman with renal disease who wishes to become pregnant?

- **Refer women who have renal disease and are planning a pregnancy** to a specialist for assessment.
    - Advise the woman to continue using effective contraception until she has been fully assessed.
    - Advise the woman not to stop taking her current medication unless otherwise directed by the specialist.
- Discuss the standard pre-conception measures.

How do I manage a woman with a history of venous thromboembolism who wishes to become pregnant?

- **Seek specialist advice from a clinician with expertise in thrombosis in pregnancy** for women who have a past history of deep vein thrombosis or pulmonary embolism, and for those with an abnormal thrombophilia screen.
- **Refer all women receiving warfarin therapy who are planning a pregnancy**to a specialist for advice, as warfarin will need to be stopped or replaced by heparin, depending on the woman's degree of risk of VTE.
- Discuss the standard pre-conception measures.

How do I manage a woman with a rheumatological condition who wishes to become pregnant?

- **Refer women with a rheumatological condition** (for example rheumatoid arthritis, systemic lupus erythematosus) who wish to become pregnant to a rheumatologist, especially if they are taking nonsteroidal anti-inflammatory drugs (NSAIDs) or disease-modifying anti-rheumatic drugs (DMARDs).
- **Ensure the use of effective contraception** whilst taking DMARDs and until a woman planning pregnancy has been reviewed by a specialist.
- Discuss the standard pre-conception measures.

How should I manage a woman with inflammatory bowel disease who wishes to become pregnant?

- **Reassure the woman** that fertility rates in women with inflammatory bowel disease (IBD) are generally similar to the rest of the population.
- **Refer women with inflammatory bowel disease who wish to become pregnant** to a gastroenterologist.
    - Advise the woman to continue using effective contraception until a full assessment by the specialist has taken place.
    - Advise the woman not to stop taking her medication unless otherwise directed by the specialist.
- Discuss the standard pre-conception measures.
    - Ensure the woman is taking an appropriate dose of folic acid.

How should I manage a woman with sickle-cell disease or thalassaemia who wishes to become pregnant?

- **Refer all women with sickle cell disease or thalassaemia to a haematologist** for assessment and monitoring.
- Women who are sickle cell or thalassaemia carriers (sickle cell or thalassaemia trait) usually have no related health issues.
    - Advise women who carry the sickle cell (S) gene that they may have problems in situations where there are changes in levels of available oxygen.
    - Advise women who are beta-thalassaemia carriers that the size of their red blood cells and levels of haemoglobin can cause confusion with iron deficiency anaemia, so testing for iron deficiency is important before taking iron supplements.
    - Advise that if their partner is also a carrier of an unusual haemoglobin gene, their baby may inherit a haemoglobin condition, therefore testing is advised.
- Ensure women with sickle cell disease, thalassaemia, or thalassaemia trait receive folic acid 5 mg daily throughout pregnancy.
- Discuss the standard pre-conception measures.

Genetic risk assessment
From age 16 years to 45 years.
How should I assess and manage a couple's risk of having a baby with an inherited genetic disorder?

- **Take a history to identify potential factors increasing genetic risks to a mother or baby.** Enquire about:
  - Family history of genetic conditions for the woman and, if possible, her partner (for example spinal muscular atrophy, cystic fibrosis, haemoglobinopathies, Fragile X syndrome, or Tay-Sachs disease).
  - Ethnic background, for example:
    - Cystic fibrosis is more common in people of Northern European descent.
    - Sickle cell disease is more common in people of African descent.
    - Alpha thalassaemia is more common in people of Southeast Asian, African, West Indian, and Mediterranean descent.
    - Beta thalassaemia is more common in people of Mediterranean, Asian, Middle Eastern, Hispanic, and West Indian descent.
    - Tay-Sachs disease is more common in people of Ashkenazi Jewish, French-Canadian, or Cajun descent.
  - Consanguinity (the couple are second cousins or closer in family relationship).
  - Obstetric history including pregnancy losses and congenital abnormalities.
- **Seek specialist advice or consider referral to a healthcare provider with genetics expertise** for couples with:
  - A personal or family history of an inherited genetic disorder.
  - A previous pregnancy affected by an inherited genetic disorder.
  - Consanguinity.
  - Recurrent pregnancy losses.

TOPIC: Antenatal care: specific points

Nausea and vomiting:

- Natural remedies - ginger and acupuncture on the 'p6' point (by the wrist) are recommended by NICE
- Antihistamines should be used first-line (BNF suggests promethazine (Phenergan) as first-line)

Vitamin D:

- NICE recommend 'All women should be informed at the booking appointment about the importance for their own and their baby's health of maintaining adequate vitamin D stores during pregnancy and whilst breastfeeding'
- 'Women may choose to take 10 micrograms (400 units) of vitamin D per day, as found in the Healthy Start multivitamin supplement'. This was confirmed in 2012 when the Chief Medical Officer advised: 'All pregnant and breastfeeding women should take a daily supplement containing 10micrograms of vitamin D, to ensure the mothers requirements for vitamin D are met and to build adequate fetal stores for early infancy'
- Particular care should be taken with women at risk (e.g. Asian, obese, poor diet)

Alcohol:

- The government now recommend pregnant women should not drink. The wording of the official advice is 'If you are pregnant or planning a pregnancy, the safest approach is not to drink alcohol at all, to keep risks to your baby to a minimum. Drinking in pregnancy can lead to long-term harm to the baby, with the more you drink the greater the risk.'

TOPIC: Antenatal care-uncomplicated pregnancy

Definition:

- **An uncomplicated pregnancy** is a singleton pregnancy where the mother is healthy and requires only routine antenatal care.
  - Women with uncomplicated pregnancies are usually managed in the community by a midwife.
- **Routine antenatal care includes:**
  - 10 antenatal care appointments with a midwife or doctor for nulliparous women.
  - 7 antenatal care appointments with a midwife or doctor for parous women.
  - An ultrasound scan between 11+2 weeks and 14+1 weeks.
  - An ultrasound scan between 18+0 weeks and 20+6 weeks.
- **Additional care is required for women:**
  - Who may be at higher risk of developing complications in pregnancy, for example those:
    - With existing medical problems such as cardiac disease (including hypertension), cystic fibrosis, renal or hepatic disease, endocrine disorders (including diabetes), psychiatric disorders, haematological disorders (for example sickle cell disease, thalassaemia or thromboembolic disorders), autoimmune conditions, epilepsy, malignancy, severe asthma or infectious disease (such as HIV or hepatitis B).
    - With a high (30 kg/m^2 or more) or low (less than 18.5 kg/m^2) body mass index (BMI).
    - Who are aged over 40 years at booking.
    - With multiple pregnancy.
    - With complex social factors including: substance abuse, recent migrants, asylum seekers or refugees, women who have difficulty reading or speaking English, young women aged under 20 years, or women who experience domestic abuse.
  - Who have experienced complications in a previous pregnancy such as pre-eclampsia, small- or large-for-gestational-age infant, stillbirth, or recurrent miscarriage.
  - Who develop complications during the current pregnancy for example placenta praevia, hypertension, gestational diabetes, abnormalities on ultrasound, or malpresentation (breech or transverse lie).
- **Additional care is led by the obstetric team working in collaboration with other relevant specialists.**

Management

From age 18 years to 40 years (Female).
Routine antenatal appointments

- **Nulliparous women should be offered 10 routine antenatal appointments with a midwife or doctor including a:**
  - Booking appointment
  - 16 weeks appointment (14–16 weeks).
  - 25 weeks appointment
  - 28 weeks appointment
  - 31 weeks appointment

    - 34 weeks appointment
    - 36 weeks appointment
    - 38 weeks appointment
    - 40 weeks appointment
    - 41 weeks appointment (for those who have not yet given birth).
- **Parous women should be offered 7 routine antenatal appointments with a midwife or doctor including a:**
    - Booking appointment
    - 16 weeks appointment (14–16 weeks).
    - 28 weeks appointment
    - 34 weeks appointment
    - 36 weeks appointment
    - 38 weeks appointment
    - 41 weeks appointment (for those who have not yet given birth).
- **All pregnant women should also be offered:**
    - An ultrasound scan appointment between 11+2 weeks and 14+1 weeks.
    - An ultrasound scan appointment between 18+0 weeks and 20+6 weeks.
- **The booking appointment should take place by 10+0 weeks of pregnancy.**
    - If the woman contacts or is referred to maternity services later than 9+0 weeks of pregnancy, she should be offered a booking appointment within 2 weeks if possible.
- **Additional or longer antenatal appointments should be offered depending on the woman's medical, social, and emotional needs.**
    - Be aware that closer monitoring and additional support may be needed for women and their babies from some minority ethnic backgrounds and those who live in deprived areas, because they are at an increased risk of adverse outcomes.
- **Management at each appointment depends on the stage of the pregnancy.**
- **Throughout pregnancy, patient information should be discussed and provided on:**
    - Physical and emotional changes during pregnancy.
    - Relationship changes during pregnancy.
    - How the woman and her partner can support each other.
    - Resources and support for expectant and new parents.
    - How the parents can bond with their baby and the importance of emotional attachment.
- **A safe environment and opportunities for the woman to discuss** concerns at home, domestic abuse, concerns about the birth (for example, if she previously had a traumatic birth), or mental health concerns should be provided at every antenatal appointment.

How should I manage a woman at first contact?

At first contact, before the booking appointment:

- **Identify women who:**
    - May need additional care — refer to a specialist as appropriate.
    - Are taking any medication (including over-the-counter drugs). If necessary, arrange for an immediate medication review.
- **Offer advice about vitamins and supplements in pregnancy, including folic acid and vitamin D.**
- **Offer advice about lifestyle factors that may affect the pregnancy, including:**
    - Diet, including how to reduce the risk of a food-acquired infection.

    - Smoking, alcohol consumption, recreational drugs use.
- **Advise the woman that if she wishes, her partner is welcome to attend antenatal appointments and classes with her for support.**

How should I manage a woman at the booking appointment?

- **Take a history asking about:**
    - General health and wellbeing including nutrition and diet, physical activity, smoking, alcohol consumption, and recreational drug use.
    - Obstetric history.
    - The medical and family history of both biological parents.
    - Previous or current mental health concerns such as depression, anxiety, severe mental illness, psychological trauma, or psychiatric treatment
        - If there are any mental health concerns, discuss with/refer to an appropriate specialist.
    - Current and recent medicines (including over-the-counter medicines, health supplements, and herbal remedies) and allergies.
    - Occupation — discuss any risks and concerns.
    - Her family and home situation including:
        - Available support network including other people who may be involved in the care of the baby.
        - Health or other issues affecting her partner or family members that may be significant for her health and wellbeing.
        - Domestic abuse, if she is alone. If not alone, ensure that there is an opportunity to have a private, one-to-one discussion at the earliest opportunity.
        - Contact details for her partner and her next of kin.
    - Any concerns that the woman (and her partner, if present) would like to discuss.
- **Assess the woman for risk of gestational diabetes:**
    - Arrange testing for gestational diabetes for women with any of the following:
        - Body mass index (BMI) above 30 kg/m^2.
        - Previous macrosomic baby weighing 4.5 kg or more.
        - Previous gestational diabetes.
        - Family history of diabetes (first degree relative with diabetes).
        - An ethnicity with a high prevalence of diabetes.
    - Consider further testing to exclude gestational diabetes in women who have:
        - Glycosuria of 2+ or above on 1 occasion.
        - Glycosuria of 1+ or above on 2 or more occasions.
    - Women diagnosed with gestational diabetes should be offered review with the joint diabetes and antenatal clinic within 1 week.
- **Assess the woman for risk of pre-eclampsia.**
    - Assess the woman's risk factors for pre-eclampsia.
    - Women at risk of pre-eclampsia should be referred for consultant-led care at booking and advised on taking aspirin.
- **Assess for risk of fetal growth restriction.**
    - Consider using guidance from the Royal College of Obstetricians and Gynaecologists
- **Assess for risk of venous thromboembolism (VTE)** at the booking appointment, and after any hospital admission or significant health event during pregnancy.
    - Risk factors include previous VTE, thrombophilia, medical comorbidities such as cancer, heart failure; active systemic lupus erythematosus, inflammatory polyarthropathy or inflammatory bowel disease; nephrotic syndrome; type 1 diabetes

mellitus with nephropathy; sickle cell disease; current intravenous drug user, obesity, older age, and smoking.
    - Pregnant women at risk of VTE should be referred to an obstetrician for further assessment and discussion on management.
- **Assess the woman's risk of and, if appropriate, discuss female genital mutilation (FGM)** in a kind, sensitive manner.
- **Offer examination and investigations including:**
    - Measuring height and weight to calculate BMI.
    - Blood pressure and urine dipstick testing to check for proteinuria.
    - Blood tests including full blood count, blood group, and rhesus D status.
- **If there are any medical concerns or review of long-term medicines is needed,** refer to/discuss with an obstetrician.
- **Discuss screening programmes in pregnancy** including risks, benefits, and limitations — advise the woman that she can accept or decline any part of any of the screening programmes offered. Offer screening for:
    - Infectious diseases in pregnancy including HIV, syphilis, and hepatitis B.
    - Sickle cell and thalassaemia.
    - Fetal anomalies.
- **Discuss changes during pregnancy, including:**
    - How the baby develops and what to expect at each stage of the pregnancy.
    - Physical, emotional, and relationship changes.
    - Support between partners.
- **Discuss staying healthy during pregnancy including:**
    - Immunization for flu, whooping cough, and other infections such as COVID-19.
    - Infections that can impact on the baby in pregnancy or during birth (such as group B streptococcus).
    - Reducing the risk of infections.
    - Safe use of medicines
    - Mental health.
    - Lifestyle including nutrition and diet, physical activity, smoking, alcohol consumption, and recreational drug use.
- **Discuss how to contact the:**
    - Midwifery team, for non-urgent advice.
    - Maternity service, for urgent concerns such as pain and bleeding.
- **Provide information on resources and support for expectant and new parents** including details of locally available antenatal classes and breastfeeding workshops.
- **Update the woman's antenatal records** with details of history, test results, examination findings, medicines, and discussions.
    - Consider reviewing the woman's previous medical records if needed, including records held by other healthcare providers.

How should I manage a nulliparous woman at 40 weeks?

- **Update the history, asking about:**
    - General health and wellbeing.
        - If the woman has had any hospital admission or significant health event since her last appointment, assess her risk of venous thromboembolism.
    - Medications (including over-the-counter medicines, health supplements, and herbal remedies).
    - Mental health — if there are any concerns, discuss with/refer to an appropriate specialist.

    - Domestic abuse if she is alone. If not alone, ensure that there is an opportunity to have a private, one-to-one discussion at the earliest opportunity.
    - Any other concerns she (and her partner, if present) would like to discuss.
- **Discuss the baby's movements** and ask if she has any concerns.
    - If there are any concerns, assess her and the baby, referring to secondary care where appropriate.
    - Advise the woman to contact maternity services at any time of day or night if she has any concerns about her baby's movements or she notices reduced fetal movements.
- **Offer examination and investigations including:**
    - Measuring blood pressure and testing for proteinuria to screen for hypertension and possible pre-eclampsia.
        - Warn the woman about the symptoms of preeclampsia and advise her to seek immediate advice if she develops any symptoms (including during the postpartum period).
    - Measure and plot symphysis–fundal height (to identify small- or large-for-gestational-age infants).
        - If there are concerns that the symphysis–fundal height is large for gestational age consider an ultrasound scan for fetal growth and wellbeing.
        - If there are concerns that the symphysis–fundal height is small for gestational age, offer an ultrasound scan for fetal growth and wellbeing, the urgency of which may depend on additional clinical findings, for example, reduced fetal movements or raised maternal blood pressure.
    - Abdominal palpation to check the position of the baby.
        - If breech presentation is suspected on abdominal palpation, offer an ultrasound scan to determine the presentation.
- **Discuss prolonged pregnancy and options on how to manage this.** Give the woman specific information on:
    - The risks associated with pregnancies that last longer than 42 weeks.
    - Options for management of prolonged pregnancy, such as a membrane sweep, induction of labour between 41+0 and 42+0 weeks, and expectant management. This may vary according to local protocols.
- **Continue discussion on:**
    - Physical, emotional, and relationship changes.
    - Sleep position.
    - Support between partners.
    - Resources for expectant and new parents.
    - Bonding with the baby and emotional attachment.
    - Preparing for labour and birth, including information about recognizing active labour, coping in labour, and creating a birth plan — confirm birth preferences.
    - The postnatal period.
- **Ensure results of any blood or screening tests from previous appointments have been reviewed, recorded, and discussed with the woman.**
    - If there are any unexpected results from examinations or investigations, offer referral according to local pathways, and provide appropriate information and support.
- **Reassess the plan of care for the pregnancy and update the woman's antenatal record** with details of history, test results, examination findings, medicines, and discussions.
    - Offer additional or longer antenatal appointments if needed, depending on the woman's medical, social, and emotional needs.

How should I manage a woman with an uncomplicated pregnancy after 41 weeks?

- **Update the history, asking about:**
    - General health and wellbeing.

        - If the woman has had any hospital admission or significant health event since her last appointment, assess her risk of venous thromboembolism.
    - Medications (including over-the-counter medicines, health supplements, and herbal remedies).
    - Mental health — if there are any concerns, discuss with/refer to an appropriate specialist.
    - Domestic abuse if she is alone. If not alone, ensure that there is an opportunity to have a private, one-to-one discussion at the earliest opportunity.
    - Any other concerns she (and her partner, if present) would like to discuss.
- **Discuss the baby's movements** and ask if she has any concerns.
    - If there are any concerns, assess her and the baby, referring to secondary care where appropriate.
    - Advise the woman to contact maternity services at any time of day or night if she has any concerns about her baby's movements or she notices reduced fetal movements.
- **Offer examination and investigations including:**
    - Measuring blood pressure and testing for proteinuria to screen for hypertension and possible pre-eclampsia.
        - Warn the woman about the symptoms of pre-eclampsia and advise her to seek immediate advice if she develops any symptoms (including during the postpartum period).
    - Measure and plot symphysis–fundal height (to identify small- or large-for-gestational-age infants).
        - If there are concerns that the symphysis–fundal height is large for gestational age consider an ultrasound scan for fetal growth and wellbeing.
        - If there are concerns that the symphysis–fundal height is small for gestational age, offer an ultrasound scan for fetal growth and wellbeing, the urgency of which may depend on additional clinical findings, for example, reduced fetal movements or raised maternal blood pressure.
    - Abdominal palpation to check the position of the baby.
        - If breech presentation is suspected on abdominal palpation, offer an ultrasound scan to determine the presentation.
- **Discuss prolonged pregnancy and options on how to manage this.** Give the woman specific information on:
    - The risks associated with pregnancies that last longer than 42 weeks.
    - Options for management of prolonged pregnancy, such as a membrane sweep, induction of labour between 41+0 and 42+0 weeks, and expectant management. This may vary according to local protocols.
- **Ensure results of any blood or screening tests from previous appointments have been reviewed, recorded, and discussed with the woman.**
    - If there are any unexpected results from examinations or investigations, offer referral according to local pathways, and provide appropriate information and support.
- **Reassess the plan of care for the pregnancy and update the woman's antenatal record** with details of history, test results, examination findings, medicines, and discussions.
    - Offer additional or longer antenatal appointments if needed, depending on the woman's medical, social, and emotional needs.

Pre-eclampsia

- **Advise all pregnant women to seek immediate medical review if they experience symptoms of pre-eclampsia (including during the first 4 weeks postpartum), such as:**
    - Severe headache.
    - Problems with vision, such as blurred vision, flashing lights, double vision, or floating spots.

    - Severe pain below the ribs.
    - Vomiting.
    - Breathlessness.
    - Sudden swelling of the face, hands, or feet.
- **Risk factors for pre-eclampsia include:**
    - Previous history of pre-eclampsia or hypertension in pregnancy.
    - Pre-existing renal disease.
    - Autoimmune disease such as systemic lupus erythematosus or antiphospholipid syndrome.
    - Type 1 or type 2 diabetes.
    - Pre-existing vascular disease, such as hypertension.
    - Nulliparity.
    - Age 40 years or older.
    - Pregnancy interval of more than 10 years.
    - Body mass index 35 kg/m^2 or above.
    - Multiple pregnancy.
    - Family history of pre-eclampsia.
- **Emergency assessment in secondary care should be arranged for any woman in whom pre-eclampsia is suspected.**

What lifestyle advice should I offer to a woman with an uncomplicated pregnancy?

What dietary advice should I offer?

- **Advise the woman to eat a variety of foods, including:**
    - Fruit and vegetables (at least 5 portions each day).
    - Carbohydrates, such as bread, pasta, rice, and potatoes.
    - Protein, such as lean meat, fish, beans, and lentils.
    - Foods rich in fibre, such as wholegrain breads, fruits, and vegetables.
    - Dairy foods, such as milk, yoghurt, and cheese.
- **Advise the woman to eat foods rich in:**
    - Folic acid such as green leafy vegetables, fortified breakfast cereals, and brown rice.
    - Vitamin D such as oily fish, eggs, fortified breakfast cereals, and fortified margarine.
- **Advise the woman that peanuts may be eaten during pregnancy (unless allergic to them).**
    - Previous advice from the Department of Health advised that pregnant women may wish to avoid eating peanuts during pregnancy. This advice has now changed because there is no clear evidence that peanuts affect the chance of the unborn baby developing a peanut allergy.
- **Advise the woman to avoid foods that may put her or her fetus at risk, including:**
    - Foods that may contain Listeria such as soft mould-ripened cheeses (Camembert, brie, and blue-veined cheese), unpasteurized milk or cheese, and pate (including vegetable pate).
        - Listeria infection in pregnancy can lead to miscarriage, stillbirth, or severe illness in the newborn baby.
    - Foods that may contain Salmonella such as uncooked or undercooked ready-prepared meals, uncooked or cured meat (including salami), and raw shellfish such as oysters.
    - Liver and liver products as these may contain high levels of vitamin A.
        - Ingesting too much vitamin A in pregnancy can increase the risk of birth defects in the baby.
        - Multivitamin supplements, fish liver oil supplements, or any other supplements containing vitamin A should be avoided in pregnancy.

    - Fish containing relatively high levels of methylmercury, such as shark, swordfish, and marlin.
        - Exposure to high levels of methylmercury in the womb can affect the nervous system of the fetus, potentially increasing the risk of learning or behavioural problems.
        - Consumption of tuna should be limited to no more than four medium-sized cans or two fresh tuna steaks per week as tuna may contain high levels of mercury.
- **Advise the woman on limiting caffeine intake:**
    - Caffeine intake should be limited to less than 200 mg a day as high levels of caffeine have been associated with low birth weight in the baby.
    - Caffeine is present in coffee, tea, chocolate, and colas. The amount of caffeine in food and drink varies, but as a guide:
        - One mug of instant coffee contains 100 mg of caffeine.
        - One mug of filter coffee contains 140 mg of caffeine.
        - One mug of tea contains 75 mg of caffeine.
        - One can of cola contains up to 40 mg of caffeine.
        - One can of 'energy' drink contains up to 80 mg of caffeine.
        - One bar of plain chocolate contains up to 50 mg of caffeine — the caffeine content of milk chocolate is about half that of plain chocolate.

What advice can I give regarding nutritional supplements?

Offer advice about vitamins and supplements in pregnancy:

- Avoid vitamin A supplementation — high levels may be teratogenic.

What advice can I give regarding exercise during pregnancy?

Advise the woman:

- **That moderate exercise may be continued or started during pregnancy.**
    - Exercise choices should reflect activity levels pre-pregnancy and should include strength training.
    - Vigorous activity is not recommended for previously inactive women.
- **To avoid:**
    - Sports that may cause abdominal trauma such as contact sports, high impact sports, and falls.
    - Scuba diving as it has been associated with an increased risk of birth defects and fetal decompression disease.

What advice can I give regarding sexual intercourse during pregnancy?

Reassure the woman that intercourse is thought to be safe during an uncomplicated pregnancy.

What advice can I give on how to avoid toxoplasmosis?

- **Toxoplasmosis is an infection with the parasite *Toxoplasma gondii*.**
    - If acquired during pregnancy, toxoplasmosis can cause stillbirth, miscarriage, intracranial anomalies, visual impairment, and developmental delay.
- **The toxoplasmosis parasite can be acquired by:**

    - Eating undercooked or uncooked meat (such as salami).
    - Eating unwashed fruit or vegetables which may be contaminated by cat faeces.
    - Accidental ingestion after handling cat litter faeces.
    - Contact with lambs or sheep.
    - Mother-to-child transmission. This occurs when primary infection occurs during pregnancy (congenital toxoplasmosis).
- **To reduce the risk of toxoplasmosis infection, advise the woman to:**
    - Wash her hands before handling food.
    - Thoroughly wash all fruit and vegetables, including ready-prepared salads, before eating.
    - Thoroughly cook raw meats and ready-prepared chilled meals.
    - Wear gloves and thoroughly wash hands after handling soil and gardening.
    - Avoid cat faeces in cat litter or in soil.
    - Avoid lambing or milking ewes and contact with newborn lambs.

What advice can I give regarding travel during pregnancy?

- **Air travel — advise the woman that:**
    - There is no evidence that air travel is harmful for healthy women with an uncomplicated pregnancy. Pregnant women with certain underlying medical conditions and those who have developed a complication during pregnancy may be advised not to fly.
    - Airlines may ask for a letter from a midwife or doctor after 27 weeks' gestation confirming the expected date of delivery and that the pregnancy is uncomplicated. Most airlines do not allow women to fly after 37 weeks.
    - Risk of developing a deep vein thrombosis (DVT) is increased while flying (due to sitting for prolonged lengths of time).
        - Risk is higher with longer flights and increases further if the woman has additional risk factors for DVT (such as a previous DVT or being overweight).
        - Individual risk should be discussed with a midwife or doctor — women with additional risk factors for DVT may be advised to have heparin injections regardless of flight duration.
- **Car travel**
    - Advise the woman to always wear her seatbelt with the diagonal strap across her body between her breasts and with the lap belt over her upper thighs. The straps should lie above and below the bump, and not over it.
- **Travelling abroad**
    - **Advise the woman that if she is travelling abroad, she should ensure that she has:**
        - Access to medical help if required.
        - Had any appropriate immunizations.
        - Adequate travel insurance.

What advice should I give about working and maternity benefits?

- **Advise that most women can continue working during pregnancy.**
    - Employers are required to assess risks which might be posed to the health and safety of pregnant women.

When are ultrasound scans offered and for what purpose?

- **During the course of an uncomplicated pregnancy, at least two ultrasound scans are usually offered:**
    - A dating scan (11+2 to 14+1 weeks). The purpose of this scan is to:
        - Determine gestational age.
        - Detect multiple pregnancies.
        - Confirm viability.
        - Detect any fetal abnormalities that might be visible early in pregnancy (such as anencephaly).
        - Provide a component of screening for Down's syndrome, Edwards' syndrome, and Patau's syndrome (if the woman chooses to be screened).
    - A fetal anomaly scan (18+0 to 20+6 weeks). The purpose of this is to locate the placenta, assess amniotic fluid, and identify 11 specified conditions that:
        - Benefit from treatment before or after birth.
        - Need treatment in a specialist setting after birth to improve health outcomes.
        - Could mean the baby may die shortly after birth.
        - Lead to a discussion about the options of continuing or terminating the pregnancy.
- **Additional ultrasound scans may be offered if:**
    - There are concerns with the progress of the pregnancy for example if symphysis–fundal height is small or large for gestational age.
    - Breech presentation is suspected (after 36+0 weeks).
    - A complication of pregnancy develops.

What advice can I give regarding the risks and limitations of a fetal anomaly scan?

- **Advise the parents that:**
    - There are no known risks of an ultrasound scan; it is thought to be safe for the mother and baby.
    - At the fetal anomaly scan, the sonographer looks specifically for 11 conditions in the baby that are associated with mortality (for example anencephaly), morbidity (for example serious heart defects), or require immediate postnatal support (for example spina bifida).
    - A fetal anomaly scan is not perfect, does not have a 100% detection rate, and inevitably some conditions will be missed or misidentified. This means that there is a chance that a baby may be born with a condition that could not be seen on scan.
    - If a screening scan identifies a potential fetal abnormality, they will be offered referral to a fetal medicine specialist for a second opinion to confirm the diagnosis.
    - Screening is their choice and they do not have to have the scan. Some people want to find out if their baby has one of the 11 physical conditions and some do not. If they choose to have the scan, they may need to make important personal decisions based on the results. Whatever they decide, they will be supported by their healthcare team.

Managing common minor ailments
From age 18 years to 40 years (Female).
Constipation

- **Give advice on:**
    - Lifestyle measures, such as adequate fiber and fluid intake and activity levels, as appropriate.
- **If lifestyle measures are ineffective, or symptoms do not respond adequately consider the need for pharmacological treatment.**

Varicose veins

- **Reassure the woman that varicose veins are common in pregnancy, are not harmful to the baby, and often improve considerably after pregnancy.**
- Consider treatment with compression stockings; these may improve the symptoms but will not prevent varicose veins from emerging.

Vaginal discharge

- **Advise pregnant women who have vaginal discharge that this is common during pregnancy, but if it is accompanied by symptoms such as itching, soreness, an unpleasant smell or pain on passing urine, there may be an infection that needs to be investigated and treated.**

Pelvic girdle pain

- **For women with pregnancy-related pelvic girdle pain,** consider referral to physiotherapy services for exercise advice and/or a non-rigid lumbopelvic belt.

Haemorrhoids

- **Offer the woman advice to minimize constipation and straining:**
    - Increasing daily fibre and fluid intake and taking regular exercise can help relieve constipation — straining during defecation should be discouraged as this can exacerbate symptoms of haemorrhoids.
- **Advise the woman about perianal hygiene,** as this may be helpful in relieving symptoms and preventing perineal dermatitis.
- **If symptoms remain troublesome, consider topical treatment:**
    - No topical haemorrhoidal preparations are licensed for use during pregnancy and women wishing to use these products should be made aware of the lack of data regarding pregnancy outcomes.
    - The potential risk of harms to the pregnant woman and/or fetus is likely to be less with simple, soothing products than with those containing corticosteroid and/or local anaesthetic (which are generally not recommended by the manufacturers for use during pregnancy due to inadequate safety evidence).

Breech presentation

- **If breech presentation is suspected based on abdominal palpation:**
    - An ultrasound scan should be performed to determine the presentation.
- **If breech presentation after 36+0 weeks is confirmed in a woman with an uncomplicated singleton pregnancy:**
    - Discuss the different options available, their benefits, risks, and implications, including:
        - External cephalic version (to turn the baby from bottom to head down).
        - Breech vaginal birth.
        - Elective caesarean birth.

TOPIC: Antenatal care: timetable

NICE issued guidelines on routine care for the healthy pregnant woman in March 2008. They recommend:

- 10 antenatal visits in the first pregnancy if uncomplicated
- 7 antenatal visits in subsequent pregnancies if uncomplicated
- Women do not need to be seen by a consultant if the pregnancy is uncomplicated

Gestation	Purpose of visit
8 - 12 weeks (ideally < 10 weeks)	Booking visit • General information e.g. diet, alcohol, smoking, folic acid, vitamin D, antenatal classes • BP, urine dipstick, check BMI Booking bloods/urine • FBC, blood group, rhesus status, red cell alloantibodies, haemoglobinopathies • Hepatitis B,C, syphilis • HIV test is offered to all women • Urine culture to detect asymptomatic bacteriuria
10 - 13+6 weeks	Early scan to confirm dates, exclude multiple pregnancy
11 - 13+6 weeks	Down's syndrome screening including nuchal scan
16 weeks	Information on the anomaly and the blood results. If Hb < 11 g/dl consider iron Routine care: BP and urine dipstick
18 - 20+6 weeks	Anomaly scan
25 weeks (only if primip)	Routine care: BP, urine dipstick, symphysis-fundal height (SFH)
28 weeks	Routine care: BP, urine dipstick, SFH Second screen for anaemia and atypical red cell alloantibodies. If Hb < 10.5 g/dl consider iron First dose of anti-D prophylaxis to rhesus negative women
31 weeks (only if primip)	Routine care as above
34 weeks	Routine care as above Second dose of anti-D prophylaxis to rhesus negative women* Information on labour and birth plan
36 weeks	Routine care as above Check presentation - offer external cephalic version if indicated Information on breast feeding, vitamin K, 'baby-blues'
38 weeks	Routine care as above
40 weeks (only if primip)	Routine care as above Discussion about options for prolonged pregnancy
41 weeks	Routine care as above Discuss labour plans and possibility of induction

TOPIC: Antenatal screening

The National Screening Committee (NSC) suggest the following policy regarding antenatal screen

Conditions which all pregnant women should be offered screening	Conditions for which screening should not be offered
Anaemia Bacteriuria Blood group, Rhesus status and anti-red cell antibodies Down's syndrome Fetal anomalies Hepatitis B,C HIV Neural tube defects Risk factors for pre-eclampsia Syphilis The following should be offered depending on the history: Placenta praevia Psychiatric illness Sickle cell disease Tay-Sachs disease Thalassaemia	Bacterial vaginosis *Chlamydia* Cytomegalovirus Fragile X Group B *Streptococcus* Toxoplasmosis

TOPIC: Gestational trophoblastic disorders

Describes a spectrum of disorders originating from the placental trophoblast:

- Complete hydatidiform mole
- Partial hydatidiform mole
- Choriocarcinoma

Complete hydatidiform mole:
Benign tumour of trophoblastic material. Occurs when an empty egg is fertilized by a single sperm that then duplicates its own DNA, hence the all 46 chromosomes are of paternal origin

Features:

- Bleeding in first or early second trimester
- Exaggerated symptoms of pregnancy e.g. hyperemesis
- Uterus large for dates
- Very high serum levels of human chorionic gonadotropin (hCG)
- Hypertension and hyperthyroidism* may be see

Management:

- Urgent referral to specialist center - evacuation of the uterus is performed
- Effective contraception is recommended to avoid pregnancy in the next 12 months

Around 2-3% go on to develop choriocarcinoma

In a **partial mole** a normal haploid egg may be fertilized by two sperms, or by one sperm with duplication of the paternal chromosomes. Therefore, the DNA is both maternal and paternal in origin. Usually triploid - e.g. 69 XXX or 69 XXY. Fetal parts may be seen

*hCG can mimic thyroid-stimulating hormone (TSH)

TOPIC: Dyspepsia - pregnancy-associated

When should I suspect dyspepsia in pregnancy?

- **Dyspepsia in pregnancy is predominantly caused by gastro-oesophageal reflux disease (GORD).** The diagnosis can be made on symptoms alone, which do not differ from the non-pregnant population.
- **Take a detailed history.**
- **Examination** is usually normal.
- **Investigations** are usually not necessary. Where they are required, they are usually carried out in secondary care and may include:
    - Manometry and pH probes.
    - An upper GI endoscopy.
    - Non-invasive testing for *Helicobacter pylori*, which may be delayed until after delivery. Evidence suggests Helicobacter pylori does not affect severity or recurrence of symptoms in pregnancy and in general eradication is not advised unless indicated by, for example, peptic ulcer disease.

Red flags

Red flag or alarm symptoms which should prompt consideration of a more serious condition or the need for urgent investigation

Differential diagnosis

Pregnancy-related disorders:

- Nausea and vomiting of pregnancy, including hyperemesis gravidarum.
- Pre-eclampsia
- HELLP syndrome (haemolysis, elevated liver enzymes, and low platelets).
- Acute fatty liver of pregnancy (rare).

Disorders unrelated to pregnancy, for example:

- Upper gastrointestinal cancer
- Peptic ulcer disease
- Oesophageal disorders, such as stricture, web or diverticulum, eosinophilic oesophagitis or oesophageal motility disorder.
- Pancreatitis
- Cholecystitis
- Irritable bowel syndrome
- Infections, including candida or acute viral hepatitis
- Cardiac disease — for example ischaemic heart disease, pericardial disease.

Management

Self-care advice

- Give lifestyle advice as first-line management.
- Advise the woman to:
    - Eat smaller meals more frequently (every 3 hours), not eat late at night (or less than 3 hours before bedtime), eat healthily, and avoid known irritants (for example alcohol, caffeine, fruit juices and carbonated drinks, chocolate, and fatty and spicy foods).
    - Keep a food diary to identify triggers.

    - Try to avoid excessive weight gain and to maintain regular physical activity.
    - Try raising the head of their bed by 10–15 cm.
    - Try to sleep on the left side rather than on the right side or supine.
    - Avoid medications that may cause or worsen symptoms, if appropriate (for example calcium-channel antagonists, anticholinergics, antidepressants, and nonsteroidal anti-inflammatory drugs).
    - Stop smoking (if applicable).
- Advise the woman to return if symptoms are not controlled by lifestyle changes, or if worsening or new symptoms develop.

Prescription

- **Antacids and alginates are recommended as first-line treatments if symptoms are not controlled adequately by lifestyle changes.**
    - Antacid preparations such as those containing combinations of aluminium and magnesium (co-magaldrox) or alginate products (for example Gaviscon® Advance) are recommended on an 'as required' basis.
    - Calcium-containing products may be used on a short-term or occasional use basis.
    - Products containing sodium bicarbonate or magnesium trisilicate are not recommended in pregnancy.
    - Liquid products are usually faster acting and more effective than tablets.
    - Advise women to avoid taking antacids within two hours of iron or folate supplements as antacids can affect absorption of other medication.
- **If symptoms are very severe, and persist despite lifestyle changes and treatment with an antacid or alginate, consider prescribing an acid-suppressing drug.**
    - CKS recommends using acid-suppressing drugs only as a last resort in severe cases in primary care, and only after a shared-care discussion about the potential risks and benefits, and about the limited evidence available regarding efficacy and safety in pregnancy. These options may be more appropriate for use in secondary care for the more serious cases or those with complications.
    - Since ranitidine has been withdrawn, the preferred H2 receptor antagonist option would be famotidine, but the manufacturer does not recommend use in pregnancy, and advises it should only be prescribed if clearly needed.
    - Of the proton pump inhibitors (PPIs) available, the manufacturer of omeprazole advises that it can be used in pregnancy but there is a lack of good quality evidence regarding safety, some possible concerns about a small increased risk of congenital malformations, and a possible association with childhood asthma and allergy. There is a lack of data for other PPIs, although the manufacturer of pantoprazole states that what data there is indicates no evidence of harm to the baby.

Contraindications and cautions associated with antacids and alginates

- Avoid antacid or alginate products if the person is known to be hypersensitive to any of the ingredients or excipients.
- For people with hepatic impairment:
    - Avoid antacids containing large amounts of sodium if the person has fluid retention.
    - Avoid antacids that cause constipation (can cause coma).
- For people with renal impairment:
    - Avoid antacids containing large amounts of sodium if the person has fluid retention.
    - Avoid, or use a lower dose of, antacids containing magnesium salts due to an increased risk of toxicity.
    - Aluminium accumulation and toxicity are a risk.

- Antacids or other agents that lower gastric acidity should not be taken within two hours of iron or folate supplements, and should be used carefully in pregnant patients with evidence of iron or other mineral deficiency because the normal absorption of minerals such as iron is enhanced by gastric acidity.

Adverse effects associated with antacids and alginates

- Side effects are rare if antacids and alginates are taken as required at the recommended dose.
- Antacids may cause either diarrhoea or constipation.
- Calcium-containing antacids can cause rebound acid secretion, hypercalcaemia and alkalosis, and milk-alkali syndrome at prolonged high doses.

Drug interactions with antacids and alginates

- Antacids affect the absorption of many drugs and may damage enteric coatings, so ideally should not be taken at the same time.

Acid-suppressing drugs

Due to the lack of good quality data and clarity regarding safety of acid-suppressing drugs in pregnancy, no specific positive recommendations are made at this time for for the use of these medications.

Referral

- Refer *immediately* (same-day) if there is dyspepsia and significant acute gastrointestinal bleeding.
- Refer *urgently* to a gastroenterologist there are features suggestive of malignancy.
    - Note that the classical alarm features of weight loss, an epigastric mass, persistent abdominal pain or iron deficiency anaemia may be difficult to interpret in pregnancy.
- Refer *non-urgently* to a gastroenterologist if:
    - Symptoms do not adequately respond to antacids, alginates, or other treatments given in primary care. (Refer more urgently if the symptoms are so severe that they affect the woman's ability to eat sufficiently.)
    - There is doubt about the diagnosis.
- Refer to an obstetrician if symptoms suggest a pregnancy-related disorder other than dyspepsia, for example pre-eclampsia or HELLP syndrome (haemolysis, elevated liver enzymes, and low platelets) — use clinical judgement regarding urgency of referral depending on the presenting problem.

TOPIC: Nausea and vomiting in pregnancy

Assessment

If a woman has a suspected diagnosis of nausea and vomiting in pregnancy:

- **Ask about:**
  - The onset, duration, frequency, and severity of symptoms, including impact on oral intake.
    - Onset of symptoms after 11 weeks of gestation usually suggests an alternative or underlying cause for symptoms, unrelated to pregnancy.
    - Dehydration may be suggested by reduced or concentrated urine output.
  - Sleep pattern in a 24-hour period.
  - The impact on her mood and emotional wellbeing, how she is coping, support available, and impact on her daily functioning and quality of life including work, home, caring for others, and social activities.
  - Additional symptoms such as weight loss, fever, urinary symptoms, headache, neurological symptoms, or abdominal pain, which may indicate an alternative cause for symptoms.
  - Any predisposing or risk factors for symptoms, such as previous history or family history.
  - Any medications prescribed or taken over-the-counter, which may be causing symptoms, such as oral iron or opioids.
  - Any co-morbidities such as diabetes mellitus or chronic kidney disease (CKD), where symptoms may increase the risk of complications such as diabetic ketoacidosis or acute kidney injury (AKI) respectively.
- **Consider using a validated questionnaire to assess the severity of symptoms,** such as the Pregnancy-Unique Quantification of Emesis (PUQE) score.
- **Examine the woman if symptoms are severe and/or affecting oral intake:**
  - Check temperature (fever may indicate an alternative cause for symptoms).
  - Check blood pressure, pulse, and assess for dehydration (suggested by dry mucous membranes, tachycardia, postural hypotension).
  - Check oxygen saturation and respiratory rate, for example if a complication is suspected.
  - Check weight to assess for weight loss and muscle wasting.
  - Perform an abdominal examination to assess for pain or tenderness (may indicate an alternative cause for symptoms).
- **Assess for clinical features suggesting an alternative or underlying cause** for symptoms, unrelated to pregnancy, and manage as appropriate.
- **Arrange investigations, depending on clinical judgement.** Additional investigations are not needed to confirm the diagnosis if there are mild and uncomplicated symptoms.
  - Dipstick the urine for ketonuria (ketones 1+ or more), if symptoms are severe and/or affecting oral intake.
  - Arrange a mid-stream urine (MSU) sample if an underlying cause, such as urinary tract infection, (UTI) is suspected.
  - Consider arranging a pelvic ultrasound scan to identify predisposing factors (for example multiple or molar pregnancy), depending on clinical judgement and local service provision. Seek specialist advice if needed.

- Note: blood tests are not routinely needed to confirm the diagnosis in primary care if there are mild, uncomplicated symptoms and an underlying cause is not suspected. Seek specialist advice if needed.

PUQE score

Table 1 shows the Pregnancy-Unique Quantification of Emesis (PUQE) score, a validated scoring system that may be used to assess the severity of symptoms of nausea and vomiting in pregnancy.

PUQE-24 scoring system					
In the last 24 hours, for how long have you felt nauseated or sick to your stomach?	Not at all (1)	1 hour or less (2)	2–3 hours (3)	4–6 hours (4)	More than 6 hours (5)
In the last 24 hours have you vomited or thrown up?	I did not throw up (1)	1-2 times (2)	3–4 times (3)	5-6 times (4)	7 or more times (5)
In the last 24 hours how many times have you had retching or dry heaves without bringing anything up?	No time (1)	1–2 times (2)	3–4 times (3)	5–6 times (4)	7 or more times (5)
Total score indicating severity of symptoms is the sum of replies to each of the three questions: mild <= 6; moderate 7–12; severe 13–15.					

Differential Diagnosis

Alternative causes for nausea and vomiting in pregnancy include:

- Genitourinary — urinary tract infection, end-stage renal disease and uraemia, pyelonephritis, ovarian torsion, renal stones.
- Metabolic/endocrine — hypercalcaemia, hyperparathyroidism, thyrotoxicosis, diabetic ketoacidosis, Addison's disease.
- Gastrointestinal — gastritis, peptic ulcer, gastroenteritis, pancreatitis, cholecystitis, cholelithiasis, bowel obstruction, hepatitis, appendicitis.
- Neurological — vestibular disease, migraine, central nervous system tumours.
- Other (pregnancy-related) — acute fatty liver of pregnancy, pre-eclampsia.
- Drug-induced — for example iron, antihypertensives, anticonvulsants, antibiotics, opioids.
- Psychological/substance misuse — for example eating disorders, alcohol/opioid/anxiolytic withdrawal.

Management

From age 13 years to 60 years (Female).

What information and advice should I offer?

If a woman has nausea and vomiting in pregnancy:

- **Reassure that mild-to-moderate symptoms are common in pregnancy** and usually resolve by 16–20 weeks of gestation.
- **Advise on the following self-care measures for mild-to-moderate symptoms:**
  - Rest as needed, and try to avoid sensory stimuli that may trigger symptoms, such as odours, heat, and noise.
  - Try eating plain biscuits or crackers in the morning.

    - Try eating bland, small, frequent protein-rich meals which are low in carbohydrate and fat.
    - Cold meals may be more easily tolerated if nausea is smell-related.
    - Drinking little and often, rather than large amounts.
    - Ginger (can be taken in fresh, tea, capsule, or syrup form).
    - Acupressure (such as over the P6 point on the ventral aspect of the wrist using a wrist band or finger pressure).
- **Advise avoiding medications that may contribute to symptoms,** such as iron-containing preparations, depending on clinical judgement.
- **Advise on the management of associated gastro-oesophageal reflux disease, oesophagitis, or gastritis symptoms.**
- **Advise on the need for urgent medical review** if the woman develops clinical features suggesting a complication or alternative cause for symptoms.
- **Advise that for any subsequent pregnancy, early use of lifestyle measures and antiemetic** drug treatment before or immediately at the start of symptoms may be helpful.

Which drug treatments should I offer?

If nausea and vomiting symptoms persist despite self-care information and advice, discuss the option of drug treatment(s), taking into account the woman's preferences, severity of symptoms, response to treatments in previous pregnancies (if appropriate), and advantages and disadvantages of different treatments.

- **Prescribe oral cyclizine or promethazine (antihistamines), prochlorperazine or chlorpromazine (phenothiazines), or the combination drug doxylamine/pyridoxine (Xonvea®) first-line,** and reassess the woman after 24 hours.
    - Consider the use of doxylamine/pyridoxine (Xonvea®, the only licensed drug treatment for this indication), depending on local prescribing guidelines.
    - If symptoms respond to treatment, continue and review the woman once a week thereafter, depending on clinical judgement.
- **If first-line treatment is ineffective, switch to a second-line antiemetic from a different drug class, such as oral metoclopramide or domperidone (dopamine receptor antagonists), or ondansetron (a 5-HT3 receptor antagonist),** and reassess the woman after 24 hours.
    - Oral metoclopramide should not be prescribed for longer than 5 days due to the risk of neurological extrapyramidal adverse effects.
    - Oral domperidone should not be prescribed for longer than 7 days due to the risk of cardiac adverse effects.
    - Oral ondansetron should not be prescribed for longer than 5 days.
        - Advise that exposure to ondansetron during the first trimester of pregnancy is associated with a small increased risk of the baby having a cleft lip and/or palate.
    - If symptoms respond to second-line treatment, continue and review the woman once a week thereafter, depending on clinical judgement.
- **If second-line treatment is ineffective, seek specialist advice.**
- **Review the need for ongoing treatment, and advise on gradually reducing and stopping medication when symptoms improve,** depending on clinical judgement.
    - It may be possible to stop antiemetic medication at around 12–16 weeks of pregnancy when symptoms have usually improved.
    - Gradually tapering the dose may reduce the risk of symptoms recurring.

When should I arrange admission or referral?

- **Consider arranging hospital admission if the woman has persistent moderate-to-severe nausea and vomiting and:**
  - Suspected hyperemesis gravidarum despite oral antiemetic treatment.
  - A suspected severe or serious complication.
  - Symptoms are not controlled with management in primary care.
  - Is unable to tolerate oral antiemetics or oral fluids.
  - Is unable to tolerate other necessary oral drug treatment, such as antibiotics for a urinary tract infection or usual medication for comorbid conditions.
  - Note: have a lower threshold for admitting to hospital or seeking specialist advice if the woman has a co-morbidity such as diabetes mellitus, which may be adversely affected by symptoms.
- **Offer referral for psychological and social support if needed.**

Cyclizine
Dose

- Prescribe cyclizine 50 mg orally up to three times a day.

Promethazine
Dose

- Promethazine is available as promethazine hydrochloride or promethazine teoclate.
  - Promethazine hydrochloride is available in two strengths — 10 mg or 25 mg tablets. Prescribe 20–25 mg as one dose at bedtime, and repeat in the morning if necessary (unlicensed indication).
  - Promethazine teoclate is available as 25 mg tablets. Prescribe 25 mg daily. The dose may be increased up to 100 mg daily if needed (unlicensed indication).
  - The Royal College of Obstetricians and Gynaecologists (RCOG) recommends use of promethazine 12.5 to 25 mg 4–8 hourly.

Prochlorperazine
Dose

- Prescribe prochlorperazine:
  - Oral tablets 5–10 mg up to three times a day. CKS notes that the Royal College of Obstetricians and Gynaecologists (RCOG) recommends oral prochloperazine 5–10 mg 6–8 hourly.
  - Buccal tablets 3–6 mg twice a day, tablets to be placed high between upper lip and gum and left to dissolve.

Chlorpromazine
Dose

Prescribe oral chlorpromazine hydrochloride:

- 10–15 mg every 4–6 hours.

Doxylamine with pyridoxine
Dose

Prescribe doxylamine with pyridoxine (Xonvea®) as a gastro-resistant tablet to people aged over 18 years:

- 20/20 mg once daily for 2 days, to be taken at bedtime; increased if necessary to 10/10 mg, to be taken in the morning and 20/20 mg, to be taken at bedtime; increased if necessary to 10/10 mg, to be taken in the morning, 10/10 mg, to be taken mid-afternoon and 20/20 mg, to be taken at bedtime; maximum 40/40 mg per day.

Contraindications and cautions

- **Prescribe doxylamine with pyridoxine (Xonvea®) with caution to people with:**
  - Asthma.
  - Bladder neck obstruction.
  - Increased intraocular pressure.
  - Narrow-angle glaucoma.
  - Pyloroduodenal obstruction.
  - Stenosing peptic ulcer.

Possible adverse effects of doxylamine with pyridoxine (Xonvea®) include:

- Dizziness, drowsiness, dry mouth, fatigue (common).
- Akathisia, anxiety, chest discomfort, constipation, diarrhoea, disorientation, dyspnoea, gastrointestinal discomfort, headaches, hyperhidrosis, irritability, malaise, palpitations, paraesthesia, skin reactions, sleep disorders, tachycardia, urinary disorders, vertigo, vision disorders (frequency unknown).

Possible drug interactions associated with doxylamine with pyridoxine (Xonvea®) include:

- Betahistine — doxylamine is predicted to decrease the effects of betahistine.
- Phenelzine — phenelzine is predicted to increase the risks of antimuscarinic adverse effects when given with doxylamine. Manufacturer advises avoid.

Metoclopramide

Dose

- For adults over 18 years of age, prescribe oral metoclopramide hydrochloride:
  - 5–10 mg up to 8-hourly, for a maximum of 5 days.

Domperidone

Dose

- For adults weighing 35 kg or more, prescribe oral domperidone:
  - 10 mg up to three times daily, for a maximum of 7 days.

Ondansetron

Dose

- For adults 18 years of age and older, prescribe oral ondansetron:
  - 4–8 mg 6–8 hourly, for a maximum of 5 days.

TOPIC: Hyperemesis gravidarum

Diagnosis:

- Persistent nausea and vomiting occurring the 1st trimester associated with a triad of > 5% pregnancy weight loss, dehydration, and electrolyte imbalance

Approach to management:

- Mild – moderate cases are treated in the primary care with antiemetics
- Severe
  - Admit to hospital if not tolerating orally
  - IV fluids (NaCl or Hartmann's solution)
  - Daily U&E; replace any K deficit
  - Keep NPO
  - Antiemetics

Line of treatment	Medication Name
First line	- H1 receptor antihistamines like cyclizine 50mg, 8 hourly PO/IM/IV
Second line	- Metoclopramide 10mg, 8 hourly PO/IM/IV
Third line	- Hydrocortisone 100mg BD IV - Prednisolone 45-50 mg PO OD if improved

- **Admission is reserved for cases that have at least one of the following:**
  - inability to keep down oral antiemetics
  - ketonuria and / or weight loss despite oral antiemetics
  - confirmed or suspected comorbidity like UTI, and inability to tolerate oral antibiotics

- **N.B:**
  - Ondansetron should no longer be used during the 1st trimester as PRAC in 2019 showed a slight increased risk of cleft lip and/or palate associated with its use
  - Women of childbearing age should consider the use of contraception

TOPIC: Bleeding in pregnancy

The table below outlines the major causes of bleeding during pregnancy. Antepartum haemorrhage is defined as bleeding after 24 weeks

1st trimester	2nd trimester	3rd trimester
Spontaneous abortion Ectopic pregnancy Hydatidiform mole	Spontaneous abortion Hydatidiform mole Placental abruption	Bloody show Placental abruption Placenta praevia Vasa praevia

Alongside the pregnancy related causes, conditions such as sexually transmitted infections and cervical polyps should be excluded.

The table below outlines the key features of each condition:

Spontaneous abortion	**Threatened miscarriage** - painless vaginal bleeding typically around 6-9 weeks **Missed (delayed) miscarriage** - light vaginal bleeding and symptoms of pregnancy disappear **Inevitable miscarriage** - complete or incomplete depending or whether all fetal and placental tissue has been expelled. **Incomplete miscarriage** - heavy bleeding and crampy, lower abdo pain. **Complete miscarriage** - little bleeding
Ectopic pregnancy	Typically, history of 6-8 weeks amenorrhoea with lower abdominal pain (usually unilateral) initially and vaginal bleeding later. Shoulder tip pain and cervical excitation may be present
Hydatidiform mole	Typically bleeding in first or early second trimester associated with exaggerated symptoms of pregnancy e.g. hyperemesis. The uterus may be large for dates and serum hCG is very high
Placental abruption	Constant lower abdominal pain and, woman may be more shocked than is expected by visible blood loss. Tender, tense uterus* with normal lie and presentation. Fetal heart may be distressed
Placental praevia	Vaginal bleeding, no pain. Non-tender uterus* but lie and presentation may be abnormal
Vasa praevia	Rupture of membranes followed immediately by vaginal bleeding. Fetal bradycardia is classically seen

*Vaginal examination should not be performed in primary care for suspected antepartum haemorrhage - women with placenta praevia may haemorrhage

TOPIC: Ectopic pregnancy

Be aware that ectopic pregnancy can present with a wide variety of symptoms and signs which may be atypical and can resemble those of other more common conditions (such as urinary tract infections and gastrointestinal conditions).

- **Suspect ectopic pregnancy in a woman of child-bearing potential who presents with any of the following:**
  - **Common symptoms:**
    - Abdominal or pelvic pain.
    - Amenorrhoea or missed period.
    - Vaginal bleeding (with or without clots).
  - **Less common symptoms:**
    - Breast tenderness.
    - Gastrointestinal symptoms (such as diarrhoea and/or vomiting).
    - Dizziness, fainting, or syncope.
    - Shoulder tip pain.
    - Urinary symptoms.
    - Passage of tissue.
    - Rectal pressure or pain on defecation.
- **Be aware that:**
  - Symptoms generally appear 6–8 weeks after the last normal menstrual period (or much later for a non-tubal ectopic pregnancy).
  - Clinical presentation can be highly variable and ranges from no symptoms to cardiovascular collapse.
- **Common signs of ectopic pregnancy include:**
  - Abdominal tenderness.
  - Pelvic tenderness.
  - Adnexal tenderness.
- **Other signs include:**
  - Cervical motion tenderness.
  - Rebound tenderness or peritoneal signs.
  - Pallor.
  - Abdominal distension.
  - Enlarged uterus.
  - Tachycardia (more than 100 beats per minute) or hypotension (less than 100/60 mmHg).
  - Shock or collapse.
  - Orthostatic hypotension.
- **Be alert for symptoms and signs of tubal rupture and intra-abdominal bleeding.** For example:
  - Vomiting and diarrhoea may be the presenting symptoms of abdominal bleeding.
  - Shoulder pain may be caused by irritation of the diaphragm due to leakage of blood from the implantation site.
  - Pallor, tachycardia, hypotension, and shock or collapse may indicate tubal rupture and severe bleeding.

Assessment

- If immediate hospital referral is not indicated confirm pregnancy with a urine pregnancy test (if not already done).
- Take a medical history, and ask about:
  - Signs and symptoms.
  - Sexual activity and use of contraception.
  - Date of the last menstrual period and when symptoms started.
    - Symptoms generally appear 6 to 8 weeks after the last normal menstrual period, but they can occur earlier, or much later in cases of non-tubal ectopic pregnancy.
  - Date of the positive pregnancy test (if one has been done).
  - Risk factors for ectopic pregnancy.
    - Exclude the possibility of ectopic pregnancy, even in the absence of risk factors, as in about a third of cases there are no identifiable risk factors.
- Perform an abdominal examination.
- Perform a gentle pelvic examination if there is no abdominal pain and tenderness.
  - Do not palpate for an adnexal or pelvic mass as this may increase the risk of rupture of an ectopic pregnancy if present.

Differential diagnosis

- **Pregnancy-related conditions that can cause bleeding in the first and second trimesters include:**
  - Miscarriage
  - Molar pregnancy — a rare condition of a slow-growing cystic tumour which develops from trophoblastic cells after fertilization. A molar pregnancy can spread to nearby tissues, which is known as an 'invasive mole', or become a malignant choriocarcinoma. This is more likely if bleeding is heavy and prolonged, symptoms of pregnancy are exaggerated, the uterus is large for dates, and vesicles or cysts are passed.
- **Pregnancy-related conditions that can cause abdominal pain in the first and second trimesters include:**
  - Miscarriage
  - Ruptured ovarian corpus luteal cyst.
  - Pregnancy-related degeneration of a fibroid.
- **Non-pregnancy-related conditions that can cause bleeding in early pregnancy include:**
  - Urethral bleeding.
  - Haemorrhoids.
  - Trauma of the cervix, vagina, or vulva.
  - Cancer of the cervix, vagina, or vulva.
  - Vaginitis.
  - Cervicitis, cervical ectropion, or cervical polyps.
- **Non-pregnancy-related conditions that can cause abdominal pain in early pregnancy include:**
  - Musculoskeletal pain.
  - Urinary tract infection.
  - Constipation.
  - Irritable bowel syndrome.
  - Diverticular disease.
  - Pelvic inflammatory disease.
  - Appendicitis.

- Renal colic.
- Bowel obstruction.
- Adhesions.
- Ovarian cyst (due to torsion, rupture, or bleeding).
- Torsion of a fibroid.
- Pelvic vein thrombosis.

Management of suspected ectopic pregnancy

From age 16 years onwards (Female).

- Refer women who are haemodynamic unstable (including pallor, tachycardia, hypotension, shock, and collapse), or in whom there is significant concern about the degree of bleeding or pain directly to Accident and Emergency.
- Use expectant management for women with a pregnancy of less than 6 weeks' gestation who are bleeding but not in pain, and who have no risk factors, such as a previous ectopic pregnancy. Advise these women:
  - To return if bleeding continues or pain develops.
  - To repeat a urine pregnancy test after 7 to 10 days and to return if it is positive.
  - That a negative pregnancy test means that the pregnancy has miscarried.
- For all women referred to an early pregnancy assessment service (or out-of-hours gynaecology service):
  - Explain the reasons for the referral and what she can expect on arrival.
  - Arrange follow up and offer appropriate support, information, and advice.
- Offer the woman the option of a follow-up appointment with a healthcare professional of her choice.

Diagnosis and treatment in secondary care

Diagnosis

- **Transvaginal ultrasound is the diagnostic tool of choice for a suspected ectopic pregnancy.** It is used to identify the location of the pregnancy and whether there is a fetal pole and heartbeat.
  - A transabdominal ultrasound may be used in some cases (for example in women with an enlarged uterus or other pelvic pathology such as fibroids or an ovarian cyst or if a transvaginal scan is unacceptable), but imaging is of poorer quality.
  - Magnetic resonance imaging (MRI) may be used as a second-line investigation tool in certain cases, for example, if the diagnosis of a caesarean scar ectopic pregnancy is equivocal and there is local expertise in MRI diagnosis.
- In women with pregnancy of unknown location (PUL, defined as a positive pregnancy test but no visible evidence of the location of the pregnancy on an ultrasound scan), measurements of serum human chorionic gonadotrophin (hCG) may be used to determine subsequent management. However, clinical symptoms are of more significance than hCG levels.
- Repeat transvaginal scans, serial hCG levels, and laparoscopy may be used to confirm the diagnosis.

Treatment

- **After an ectopic pregnancy has been confirmed, treatment options include expectant management (watchful waiting), medical management, and surgery.**
  - The choice of treatment will depend on factors such as the haemodynamic stability of the woman, the site of implantation of the ectopic pregnancy, the risk of tubal rupture,

serum hCG level, the level of pain the woman has, and the acceptability of the method of treatment to the woman.

- **Expectant management** is an option for a few women who are clinically stable and pain free. Active intervention will be considered if symptoms of ectopic pregnancy occur or if levels of serum hCG fail to decrease at an acceptable rate.
- **Medical management** involves the use of drug treatment, most commonly parenteral methotrexate. It is offered first line to women who are able to return for follow up and who have all of the following:
    - No significant pain.
    - An unruptured ectopic pregnancy with an adnexal mass smaller than 35 mm with no visible heartbeat.
    - Serum hCG level less than 1500 IU/L.
    - No intrauterine pregnancy (as confirmed on an ultrasound scan).
- **Surgery,** by means of salpingectomy or salpingotomy, is performed laparoscopically or by open surgery. It is offered first line to women who are unable to return for follow up after methotrexate treatment or who have an ectopic pregnancy and any of the following:
    - Significant pain.
    - Adnexal mass of 35 mm or larger.
    - Fetal heartbeat visible on an ultrasound scan.
    - Serum hCG level of 5000 IU/L or more.
- **A choice of either methotrexate or surgical management** is offered to women with an ectopic pregnancy who have a serum hCG level of at least 1500 IU/L and less than 5000 IU/L, are able to return for follow up, and meet all of the following criteria:
    - No significant pain.
    - An unruptured ectopic pregnancy with an adnexal mass smaller than 35 mm with no visible heartbeat.
    - No intrauterine pregnancy (as confirmed on an ultrasound scan).
- All women (except those who have had a salpingectomy) will be followed up according to local protocols to ensure that serum hCG levels decrease at an acceptable rate until non-pregnant levels are reached.
- **Anti-D immunoglobulin** is offered to all rhesus-negative women who have had surgical removal of an ectopic pregnancy.

Follow up after an ectopic pregnancy

From age 16 years onwards (Female).

- **Ensure that arrangements for routine antenatal care are cancelled (if they have begun).**
- **Give the woman the opportunity to discuss any questions she has about the ectopic pregnancy,** including the risk of recurrence and the effect on future fertility.
- **Assess the woman's psychological wellbeing** and offer counselling if appropriate.
    - Be aware that:
        - Pregnancy loss, irrespective of gestation or location of pregnancy is a bereavement, and while the emotional trauma experienced is variable, the potential adverse impact should not be underestimated.
        - The psychological impact of early pregnancy loss may seriously affect a significant proportion of women, their partners and families.
        - Women experience high levels of post-traumatic stress, anxiety, and depression after early pregnancy loss. Distress declines over time but can remain at clinically significant levels at 9 months.
- **Advise women treated with methotrexate to wait at least 3 months before trying to conceive again.**

    - Women who have been treated with methotrexate should be advised that effective contraception is recommended during and for at least 3 months after treatment in view of the teratogenic effects of methotrexate.
- **Give appropriate contraception advice,** including that:
    - Any method of contraception can be safely used after ectopic pregnancy.
        - A history of ectopic pregnancy is not a contraindication to any form of hormonal or intrauterine contraception.
    - Additional contraceptive precautions (e.g. barrier methods/abstinence) are required if hormonal contraception is started 5 days or more after surgical treatment or administration of methotrexate for ectopic pregnancy. Additional contraceptive precaution is not required if contraception is initiated immediately or within 5 days of treatment of ectopic pregnancy.
    - That intrauterine contraception (IUC) is one of the most effective methods of contraception and so the absolute risk of any pregnancy including ectopic pregnancy is extremely low.
    - If pregnancy occurs with IUC in situ, or after tubal occlusion, there is an increased risk of ectopic pregnancy and therefore the location of the pregnancy should be confirmed by ultrasound as soon as possible.
- **Ensure that all rhesus-negative women who have had surgical removal of an ectopic pregnancy have received anti-D immunoglobulin.**
    - Be aware that anti-D immunoglobulin should not be given to women who:
        - Receive solely medical management for an ectopic pregnancy.
        - Have a pregnancy of unknown location.
- **Advise the woman that she should inform her GP as soon as possible about any future pregnancy** so that an ultrasound scan can be arranged at 6–7 weeks to establish the location and viability of the pregnancy.

TOPIC: Miscarriage

Definition

- **Miscarriage is the spontaneous loss of pregnancy before the fetus reaches viability.** The term includes all pregnancy losses from the time of conception until 24 weeks of gestation.
- **A miscarriage is described as:**
  - Early — if it occurs before 13 weeks of gestation.
  - Late — if it occurs between 13 and 24 weeks of gestation.
- **The stages of miscarriage may include:**
  - Threatened miscarriage — when vaginal bleeding, with or without lower abdominal pain, occurs in the first 24 weeks of gestation. Pregnancy may continue.
  - Inevitable miscarriage — when specific clinical features indicate that a pregnancy is in the process of physiological expulsion from the uterine cavity. Pregnancy will not continue and will proceed to incomplete or complete miscarriage.
  - Missed miscarriage (also known as delayed or silent miscarriage) — when a non-viable pregnancy is identified on an ultrasound scan without associated pain and bleeding.
  - Incomplete miscarriage — when products of conception are partially expelled from the uterus. Many incomplete miscarriages may be missed miscarriages.
  - Complete miscarriage — when all the products of conception have been expelled from the uterus, and bleeding has stopped.
- **Recurrent miscarriage is the loss of three or more pregnancies before 24 weeks of gestation.**
  - In some European countries and the US, recurrent miscarriage is defined as two or more miscarriages, either consecutive or non-consecutive.

Prevalence

- **Miscarriage is the most common cause of pregnancy loss and one of the most common complications in early pregnancy.**

Causes

- **The most common cause of spontaneous pregnancy loss in the first trimester is fetal chromosomal abnormalities** (detected in 50–85% of pregnancy tissue specimens after spontaneous miscarriage).
- **No causes or associations are found in about 50% of couples with recurrent miscarriage (unexplained or idiopathic recurrent miscarriage).**
  - Up to 75% of these couples achieve a successful live birth in future pregnancies if given only supportive care and psychological support in a dedicated early pregnancy assessment unit (EPAU).

Risk factors

- **Risk factors for miscarriage include:**
  - Advanced maternal age — advanced maternal age (usually defined as age 35 years or older) is strongly related to fetal chromosomal abnormalities.
  - Advanced paternal age

- Congenital uterine anomalies — there is an increased risk of miscarriage in women with septate or bicornuate uteri.
- Endocrine disorders — maternal endocrine disorders, such as polycystic ovary syndrome (PCOS), diabetes mellitus, and thyroid disease, have been associated with miscarriage. Poorly controlled diabetes with a high HbA1c in the first trimester may increase the risk of miscarriage and fetal malformation.
- Vitamin D deficiency.
- Previous miscarriage — the risk of a future miscarriage increases with increasing number of previous miscarriages.
- Black ethnic background
- Maternal lifestyle choices — there is some evidence for an association between sporadic miscarriage and smoking, alcohol consumption (approximately 10 units per week), increased caffeine intake, and obesity. Exposure to environmental risk factors (such as air pollution and household chemicals) may increase the risk of sporadic miscarriage.

- **Risk factors for recurrent miscarriage may include:**
  - Chromosomal abnormalities — chromosome abnormalities are the most common cause of explained recurrent miscarriage. However, the frequency of fetal chromosomal abnormality significantly decreases with increasing number of previous miscarriages.
  - Blood clotting factors — antiphospholipid syndrome is found in about 15% of people with recurrent miscarriage. Inherited thrombophilic defects (such as factor V Leiden mutation and hyperhomocysteinaemia) predispose people to thromboembolic events; this may partially explain the weak association of some of these defects with recurrent miscarriage.
  - Parental chromosomal anomaly — these account for about 3–5% of cases of recurrent miscarriage and are most commonly balanced reciprocal or Robertsonian translocations.
  - PCOS — the risk of recurrent miscarriage is thought to be increased by hyperandrogenism, obesity, and insulin resistance.
  - Male factor — meta-analyses have found an association between increased sperm DNA fragmentation in male partners and unexplained recurrent miscarriage.
  - Subclinical hypothyroidism and thyroid autoantibodies — uncontrolled thyroid disorders are associated with miscarriages. The presence of thyroid antibodies is associated with a higher miscarriage rate; however, the presence of association does not mean causation and could be explained by mechanisms such as an underlying autoimmune state or mild thyroid failure.
  - Prolactin imbalances — both increased and reduced levels of prolactin have been implicated in recurrent miscarriage. Overall, it appears that maintaining a normal level of prolactin may be beneficial in reducing the risk of recurrent miscarriage.
  - Body mass index (BMI) — evidence from a systematic review and meta-analysis of lifestyle factors and risk of recurrent pregnancy loss showed that women with a BMI below 19 kg/m^2 and above 25 kg/m^2 were at higher odds of recurrent miscarriage.

Complications

- **Complications of miscarriage include:**
  - Psychological distress — the loss of an early pregnancy can affect couples as significantly as a neonatal death
  - Infections and sepsis — these may occur before or after the miscarriage.
  - Treatment complications, including:

        - Bleeding — retained early pregnancy tissue within the uterine cavity could result in persistent vaginal bleeding and suprapubic pain. Brisk bleeding may occur after complete evacuation of the uterus.
        - Uterine or cervical perforation — may occur during surgical evacuation of the uterus and can result in severe blood loss and shock.
        - Asherman's syndrome (rare) — may occur as a result of trauma to the endometrial lining during a curettage procedure. Scar tissue forms inside the uterus and/or the cervix and can result in infertility, recurrent miscarriage, and high-risk pregnancies.
    - Recurrent miscarriage — the risk of a future miscarriage increases with increasing number of previous miscarriages.

Assessment

- **Confirm pregnancy with a urine pregnancy test.**
    - If the woman has had a recent home pregnancy test, consider repeating the test to reconfirm the diagnosis of pregnancy.
- **Take a detailed history.**
    - Ask about her presenting symptoms, including when they started, as well as the duration, severity, and nature.
        - Suspect a miscarriage if a woman who is pregnant, or has symptoms of pregnancy (such as amenorrhoea or breast tenderness), presents with vaginal bleeding, with or without pain, in the first 24 weeks of pregnancy.
        - Bleeding is typically scanty, varying from a brownish discharge to bright red bleeding, and may recur over several days.
        - Lower abdominal cramping pain or lower backache, when it occurs, usually develops after the onset of bleeding.
    - Ask about her:
        - Gynaecological history, including menstrual cycle (last menstrual period [LMP], cycle length, and nature of bleeding), cervical smear history, and previous gynaecological diagnosis or surgery.
        - Obstetric history, including previous live births, method of delivery, and previous miscarriage, ectopic pregnancy, or termination.
        - Medical history, including a history of endocrine disorders (such as diabetes, polycystic ovary syndrome, and thyroid disorder).
        - Social history, including smoking and alcohol status.
    - Ask about symptoms (or risk factors) of ectopic pregnancy.
        - Common symptoms include abdominal or pelvic pain, amenorrhoea or missed period, and vaginal bleeding with or without clots.
        - Other reported symptoms include breast tenderness; gastrointestinal symptoms; dizziness, fainting, or syncope; shoulder tip pain; urinary symptoms; passage of tissue; and rectal pressure or pain on defecation.
        - Exclude the possibility of ectopic pregnancy, even in the absence of risk factors (such as previous ectopic pregnancy).
        - Be aware that atypical presentations for ectopic pregnancy are common.
- **Examine the woman.**
    - Assess for signs of haemodynamic instability, including pallor, tachycardia, tachypnoea, and hypotension.
    - Consider performing a speculum examination to assess the cervical os, rule out other sources of bleeding (such as an ectropion, a cervical polyp, or a genital tract laceration), quantify the bleeding, and assess for visible products of conception.
    - Examine for signs of an ectopic pregnancy.

        - Perform an abdominal examination to assess for signs of an acute abdomen (for example, rebound tenderness and guarding), which may be suggestive of an ectopic pregnancy.
        - If there is no abdominal pain or tenderness, perform a gentle pelvic examination. Do not palpate for an adnexal or pelvic mass, as this may increase the risk of rupture of an ectopic pregnancy if present.
        - Common signs of an ectopic pregnancy include pelvic tenderness, adnexal tenderness, and abdominal tenderness.
        - Other reported signs include cervical motion tenderness, rebound tenderness or peritoneal signs, pallor, abdominal distension, enlarged uterus, tachycardia (more than 100 beats per minute), hypotension (less than 100/60 mmHg), shock or collapse, and orthostatic hypotension.
        - Be aware that the clinical features of ectopic pregnancy can resemble those of other conditions, such as gastrointestinal conditions or urinary tract infections.
- **Be alert for symptoms and signs of tubal rupture and intra-abdominal bleeding.** For example:
    - Vomiting and diarrhoea may be the presenting symptoms of abdominal bleeding.
    - Shoulder pain may be caused by irritation of the diaphragm due to leakage of blood from the implantation site.
    - Pallor, tachycardia, hypotension, and shock or collapse may indicate tubal rupture and severe bleeding.
- Exclude other conditions that can cause bleeding or pain in pregnancy, such as haemorrhoids and adnexal torsion.

Differential diagnosis

- **Other possible causes of bleeding include:**
    - Pregnancy-related conditions, such as:
        - Ectopic pregnancy.
        - Molar pregnancy — a rare condition of a slow-growing cystic tumour which develops from trophoblastic cells after fertilization. This is more likely if bleeding is heavy and prolonged, symptoms of pregnancy are exaggerated, the uterus is large for dates, and/or vesicles or cysts are passed.
    - Non-pregnancy-related conditions, such as:
        - Cervicitis, cervical ectropion, or cervical polyps.
        - Cancer of the cervix, vagina, or vulva.
        - Haemorrhoids.
        - Trauma of the cervix, vagina, or vulva.
        - Urethral bleeding.
        - Vaginitis.
- **Other possible causes of abdominal pain include:**
    - Pregnancy-related conditions, such as:
        - Ruptured ovarian corpus luteal cyst.
        - Adnexal torsion.
        - Pregnancy-related degeneration of a fibroid.
    - Non-pregnancy-related conditions, such as:
        - Musculoskeletal pain.
        - Urinary tract infection.
        - Constipation.
        - Irritable bowel syndrome.
        - Pelvic inflammatory disease.

        - Appendicitis.
        - Renal colic.
        - Bowel obstruction.
        - Adhesions.
        - Ovarian cyst (due to torsion, rupture, or bleeding).
        - Torsion of a fibroid.
        - Pelvic vein thrombosis.

Managing suspected first trimester miscarriage

From age 16 years onwards (Female).

- **Arrange immediate hospital admission if:**
    - The woman has signs of haemodynamic instability, including pallor, tachycardia, tachypnoea, hypotension, shock, and collapse. Resuscitate with intravenous fluids if available.
    - There is significant concern about the degree of bleeding or pain.
- **Arrange immediate admission to an early pregnancy assessment unit (EPAU) or out-of-hours gynaecology service for:**
    - Women with a positive pregnancy test and one or more of the following on examination:
        - Abdominal pain and tenderness.
        - Pelvic tenderness.
        - Cervical motion tenderness.
- **Refer to EPAU or out-of-hours gynaecology service (with urgency dependent on the clinical situation):**
    - Women with bleeding or other symptoms and signs of early pregnancy complications who have one or more of the following:
        - Pain.
        - A pregnancy of 6 weeks gestation or more.
        - A pregnancy of uncertain gestation.
- **Use expectant management for women with pregnancy of less than 6 weeks gestation who are bleeding but not in pain and who have no risk factors, such as previous ectopic pregnancy.**

    - Advise these women:

        - To repeat a urine pregnancy test after 7–10 days and return if it is positive.
        - That a negative pregnancy test means that the pregnancy has miscarried.
        - To return if bleeding continues or pain develops.
    - Refer to EPAU or out-of-hours gynaecology service:
        - Women who return with worsening symptoms. The decision on whether she should be seen immediately or within 24 hours will depend on the clinical situation.
        - If there is doubt about the viability of a pregnancy.

Diagnosis and management in secondary care

Diagnosis

- Ultrasonography is used to assess the location and viability of the pregnancy.

    - A transvaginal ultrasound scan is preferred because, compared with a transabdominal scan, it provides good-quality imaging and can offer clearer pictures of the womb, ovaries, and surrounding areas.
    - A transabdominal ultrasound scan may be considered:
        - For women with an enlarged uterus or other pelvic pathology, such as fibroids or an ovarian cyst.
        - If a transvaginal ultrasound scan is unacceptable to the woman.
- A single transvaginal ultrasound scan may not always accurately diagnose miscarriage. Therefore, women with a suspected miscarriage are usually offered a second assessment to confirm the diagnosis.

Expectant management

- Expectant management is the first-line management option for women with a confirmed diagnosis of miscarriage.
- Other management options (medical or surgical management) may be considered if:
    - The woman is at increased risk of haemorrhage (for example, she is in the late first trimester).
    - The woman has previous adverse and/or traumatic experiences associated with pregnancy (for example, stillbirth, miscarriage or antepartum haemorrhage).
    - The woman is at increased risk from the effects of haemorrhage (for example, if she has coagulopathies or is unable to have a blood transfusion).
    - There is evidence of infection.
    - Expectant management is not acceptable to the woman.
- The National Institute for Health and Care Excellence (NICE) recommends that:
    - If the resolution of bleeding and pain indicates that the miscarriage has completed during 7–14 days of expectant management, the woman should be advised to do a pregnancy test after 3 weeks and return if it is positive.
    - A repeat scan should be offered if, after the period of expectant management, the bleeding and pain:
        - Has not started (suggesting that the process of miscarriage has not begun).
        - Are persisting and/or increasing (suggesting incomplete miscarriage).

Medical management

- Medical management is offered if expectant management is not clinically appropriate or a woman has ongoing symptoms after 14 days of expectant management.
- For the medical management of missed miscarriage, the woman is offered 200 mg oral mifepristone and, 48 hours later, 800 micrograms of misoprostol (vaginal, oral, or sublingual) unless the gestational sac has already been passed.
- For the medical management of incomplete miscarriage, the woman is offered a single dose of misoprostol 600 micrograms (vaginal, oral, or sublingual). Misoprostol 800 micrograms can be used as an alternative to allow alignment of treatment protocols for both missed and incomplete miscarriages.
- The woman is advised to do a pregnancy test after 3 weeks and return if it is positive.

Surgical management

- Surgical intervention may be required if products of conception are retained despite medical treatment or offered if the woman has ongoing symptoms after 14 days of expectant management.

- The women are offered a choice of manual vacuum aspiration under local anaesthetic or surgical management under general anaesthetic.

Anti-D immunoglobulin

- Anti-D immunoglobulin prophylaxis is offered to all rhesus-negative women who have had a surgical procedure to manage a miscarriage.
- It should not be given to women who:
    - Receive solely medical management for an ectopic pregnancy or miscarriage.
    - Have a threatened miscarriage.
    - Have had a complete miscarriage.
    - Have a pregnancy of unknown location.

Follow up after a miscarriage
From age 16 years onwards (Female).

- **After a miscarriage:**
    - Ensure that arrangements for routine antenatal care are cancelled (if they have been started).
    - Assess the woman's psychological well-being. Offer counselling if appropriate.
    - Ensure that all rhesus-negative women who have had a surgical procedure to manage miscarriage have received anti-D immunoglobulin prophylaxis.
- **Discuss any questions the woman has about her miscarriage, including:**

    - The causes and risk factors of miscarriage.
    - When to resume sexual activity.
        - Advise avoidance of sexual intercourse until miscarriage symptoms have completely settled.
    - When to try for another baby.
        - Explain that menstruation can be expected to resume within 4–8 weeks of the miscarriage, and ovulation will occur before this.
        - For women who wish to become pregnant, advise that they can do so as soon as they feel psychologically and physically ready, and offer pre-conception advice.
        - For women who do not wish to become pregnant, advise the use of contraception immediately after the miscarriage.

Managing recurrent miscarriage
From age 16 years onwards (Female).
For women with recurrent miscarriage (three or more first-trimester miscarriages):

- **Offer referral for investigation and management.**
    - Advise that it may not be possible to determine the cause of recurrent miscarriage.
        - If no cause is found, the prognosis for a successful future pregnancy with supportive care alone is about 75%. However, the prognosis worsens with increasing maternal age and the number of previous miscarriages.
- **Following referral, the woman may be offered:**
    - Testing for acquired thrombophilia, particularly for lupus anticoagulant and anticardiolipin antibodies, before pregnancy. If antiphospholipid antibodies are found, treatment with aspirin plus heparin until at least 34 weeks of gestation will be considered in future pregnancies.

- Cytogenetic analysis on pregnancy tissue of the third and subsequent first-trimester miscarriage(s).
- Parental peripheral blood karyotyping (if testing of pregnancy tissue reports an unbalanced structural chromosomal abnormality or there is unsuccessful or no pregnancy tissue available for testing).
- Assessment for congenital uterine anomalies.
- Thyroid function tests and assessment for thyroid peroxidase (TPO) antibodies.
    - Thyroxine supplementation may be considered for women with moderate subclinical hypothyroidism (thyroid-stimulating hormone [TSH] more than 4 mIU/l) but is not routinely recommended for women with mild subclinical hypothyroidism (TSH more than 2.5 mIU/l) irrespective of TPO status.
    - Regular TSH measurement from 7–9 weeks of gestation is recommended in cases with TPO and/or SCH.
- Lifestyle advice, including (as appropriate) maintaining a body mass index (BMI) between 19 kg/m^2 and 25 kg/m^2, smoking cessation, limiting alcohol consumption, and limiting caffeine to less than 200 mg/day.

- **If a woman experiences further miscarriage(s) following referral,** ensure that the couple are offered supportive care, ideally in the setting of a dedicated recurrent miscarriage clinic.

TOPIC: Placenta praevia

Placenta praevia describes a placenta lying wholly or partly in the lower uterine segment

Epidemiology:

- 5% will have low-lying placenta when scanned at 16-20 weeks-gestation
- Incidence at delivery is only 0.5%, therefore most placentas rise away from the cervix

Associated factors:

- Multiparity
- Multiple pregnancy
- Embryos are more likely to implant on a lower segment scar from previous caesarean section

Clinical features:

- Shock in proportion to visible loss
- No pain
- Uterus not tender
- Lie and presentation may be abnormal
- Fetal heart usually normal
- Coagulation problems rare
- Small bleeds before large

Diagnosis:

- Digital vaginal examination should not be performed before an ultrasound as it may provoke a severe haemorrhage
- Placenta praevia is often picked up on the routine 20 weeks abdominal ultrasound
- The RCOG recommend the use of transvaginal ultrasound as it improves the accuracy of placental localisation and is considered safe

Classical grading:

- I - placenta reaches lower segment but not the internal os
- II - placenta reaches internal os but doesn't cover it
- III - placenta covers the internal os before dilation but not when dilated
- IV ('major') - placenta completely covers the internal os

TOPIC: Placental abruption

Placental abruption describes separation of a normally sited placenta from the uterine wall, resulting in maternal haemorrhage into the intervening space

Epidemiology:

- Occurs in approximately 1/200 pregnancies

Cause - not known but associated factors:

- Proteinuric hypertension
- Cocaine use
- Multiparity
- Maternal trauma
- Increasing maternal age

Clinical features:

- Shock out of keeping with visible loss
- Pain constant
- Tender, tense uterus
- Normal lie and presentation
- Fetal heart: absent/distressed
- Coagulation problems
- Beware pre-eclampsia, DIC, anuria

TOPIC: Gestational diabetes

Gestational diabetes is the second most common medical disorder complicating pregnancy (after hypertension), affecting around 4% of pregnancies.

Risk factors for gestational diabetes:

- BMI of > 30 kg/m²
- previous macrosomic baby weighing 4.5 kg or above
- previous gestational diabetes
- first-degree relative with diabetes
- family origin with a high prevalence of diabetes (South Asian, black Caribbean and Middle Eastern)

Screening for gestational diabetes (ADA 2021)

- One step strategy
  - Perform a 75-g OGTT with plasma glucose measurement when patient is fasting at 1 and 2hr, at 24-28 weeks of gestation for women not previously diagnosed with diabetes.
  - The OGTT should be performed in the morning after an overnight fast of at least 8 hours.
  - The diagnosis of GDM is made when any of the following plasma glucose values are met or exceeded:
    - Fasting: 5.1 mmol/L
    - 1hr: 10.0 mmol/L
    - 2hr: 8.5 mmol/L
- Consider screening before 15 weeks of gestation or at the first prenatal visit for those with signs of insulin resistance or conditions associated with insulin resistance (acanthosis nigricans, hypertension, dyslipidemia, polycystic ovarian syndrome, or small for gestational age birthweight)
- HbA1c is not used in monitoring or screening of gestational diabetes mellitus due to the physiological increase in red blood cell turnover during pregnancy.

Management of gestational diabetes (ADA 2021):

- newly diagnosed women should be seen in a joint diabetes and antenatal clinic within a week
- women should be taught about self-monitoring of blood glucose
- advice about diet (including eating foods with a low glycaemic index) and exercise should be given
- Lifestyle behavior change is an essential component of management of gestational diabetes mellitus of many women. Insulin should be added if needed to achieve glycemic targets.
- Insulin is the preferred medication for treating hyperglycemia in gestational diabetes mellitus.
- Metformin and glyburide should not be used as first-line agents, as both cross the placenta to the fetus.
- Other oral and non-insulin injectable glucose-lowering medications lack long-term safety data.
- Metformin, when used to treat polycystic ovarian syndrome and induce ovulation, should be discontinued by the end of first trimester.

Management of pre-existing diabetes:

- weight loss for women with BMI of > 27 kg/m^2
- stop oral hypoglycaemic agents, apart from metformin, and commence insulin
- folic acid 5 mg/day from pre-conception to 12 weeks-gestation
- detailed anomaly scan at 20 weeks including four-chamber view of the heart and outflow tracts
- tight glycaemic control reduces complication rates
- treat retinopathy as can worsen during pregnancy

Glycemic Targets in Pregnancy (ADA 2021):

- Fasting glucose <5.3 mmol/L
- 1 hour post prandial <7.8 mmol/L
- 2 hour-post prandial <6.7 mmol/L

Complications of diabetes in pregnancy

Maternal complications:

- polyhydramnios - 25%, possibly due to fetal polyuria
- preterm labour - 15%, associated with polyhydramnios

Neonatal complications:

- macrosomia (although diabetes may also cause small for gestational age babies)
- hypoglycaemia (secondary to beta cell hyperplasia)
- respiratory distress syndrome: surfactant production is delayed
- polycythaemia: therefore, more neonatal jaundice
- malformation rates increase 3-4 folds e.g. sacral agenesis, CNS and CVS malformations (hypertrophic cardiomyopathy)
- stillbirth
- hypomagnesaemia
- hypocalcaemia
- shoulder dystocia (may cause Erb's palsy)

Post-partum care (ADA 2021):

- Women with a recent history of gestational diabetes mellitus should be screened at 4-12 weeks postpartum using the 75-g oral glucose tolerance test and clinically appropriate non pregnancy diagnostic criteria.
- Women with a history of gestational diabetes mellitus found to have prediabetes should receive intensive lifestyle interventions and/or metformin to prevent diabetes.
- Women with a history of gestational diabetes mellitus should have lifelong screening for the development of type 2 diabetes or prediabetes every 1-3 years.
- Women with a history of gestational diabetes mellitus should seek preconception screening for diabetes and preconception care to identify and treat hyperglycemia and prevent congenital malformations.
- Postpartum care should include psychosocial assessment and supportive self-care.

TOPIC: Hypertension in pregnancy

NICE published guidance in 2010 on the management of hypertension in pregnancy. They also made recommendations on reducing the risk of hypertensive disorders developing in the first place. Women who are at high risk of developing pre-eclampsia should take aspirin 75mg od from 12 weeks until the birth of the baby. High risk groups include:

- hypertensive disease during previous pregnancies
- chronic kidney disease
- autoimmune disorders such as SLE or antiphospholipid syndrome
- type 1 or 2 diabetes mellitus

The classification of hypertension in pregnancy is complicated and varies. Remember, in normal pregnancy:

- blood pressure usually falls in the first trimester (particularly the diastolic), and continues to fall until 20-24 weeks
- after this time the blood pressure usually increases to pre-pregnancy levels by term

Hypertension in pregnancy is usually defined as:

- systolic > 140 mmHg or diastolic > 90 mmHg
- or an increase above booking readings of > 30 mmHg systolic or > 15 mmHg diastolic

After establishing that the patient is hypertensive, they should be categorised into one of the following groups

Pre-existing hypertension	Pregnancy-induced hypertension (PIH, also known as gestational hypertension)	Pre-eclampsia
A history of hypertension before pregnancy or an elevated blood pressure > 140/90 mmHg before 20 weeks gestation No proteinuria, no oedema Occurs in 3-5% of pregnancies and is more common in older women	Hypertension (as defined above) occurring in the second half of pregnancy (i.e. after 20 weeks) No proteinuria, no oedema Occurs in around 5-7% of pregnancies Resolves following birth (typically after one month). Women with PIH are at increased risk of future pre-eclampsia or hypertension later in life	Pregnancy-induced hypertension in association with proteinuria (> 0.3g / 24 hours) Oedema may occur but is now less commonly used as a criteria. Occurs in around 5% of pregnancies

TOPIC: Pre-eclampsia

Pre-eclampsia describes the emergence of high blood pressure during pregnancy that may be a precursor to a woman developing eclampsia and other complications. **It is classically a triad of 3 things:**

- new-onset hypertension
- proteinuria
- oedema

Definition:

- new-onset blood pressure ≥ 140/90 mmHg after 20 weeks of pregnancy, AND 1 or more of the following:
  - proteinuria
  - other organ involvement (see list below for examples): e.g. renal insufficiency (creatinine ≥ 90 umol/L), liver, neurological, haematological, uteroplacental dysfunction

Features:

- eclampsia
  - other neurological complications include altered mental status, blindness, stroke, clonus, severe headaches or persistent visual scotomata
- fetal complications
  - intrauterine growth retardation
  - prematurity
- liver involvement (elevated transaminases)
- haemorrhage: placental abruption, intra-abdominal, intra-cerebral
- cardiac failure

Features of severe pre-eclampsia:

- hypertension: typically, > 160/110 mmHg and proteinuria as above
- proteinuria: dipstick ++/+++
- headache
- visual disturbance
- papilloedema
- RUQ/epigastric pain
- hyperreflexia
- platelet count < 100 * 10^6/l, abnormal liver enzymes or HELLP syndrome

Prevention:
NICE divide risk factors into high and moderate risk:

High risk factors	Moderate risk factors
• hypertensive disease in a previous pregnancy • chronic kidney disease • autoimmune disease, such as systemic lupus erythematosus or antiphospholipid syndrome • type 1 or type 2 diabetes	• first pregnancy • age 40 years or older • pregnancy interval of more than 10 years

High risk factors	Moderate risk factors
• chronic hypertension	• body mass index (BMI) of 35 kg/m² or more at first visit • family history of pre-eclampsia • multiple pregnancy

Reducing the risk of hypertensive disorders in pregnancy:

- women with the following should take aspirin 75-150mg daily from 12 weeks-gestation until the birth
  - ≥ 1 high risk factors
  - ≥ 2 moderate factors

Management:

Initial assessment

- NICE recommend arranging emergency secondary care assessment for any woman in whom pre-eclampsia is suspected
- women with blood pressure ≥ 160/110 mmHg are likely to be admitted and observed

Further management

- oral labetalol is now first-line following the 2010 NICE guidelines. Nifedipine (e.g. if asthmatic) and hydralazine may also be used
- delivery of the baby is the most important and definitive management step. The timing depends on the individual clinical scenario

TOPIC: Pregnancy: jaundice

Intrahepatic cholestasis of pregnancy

Intrahepatic cholestasis of pregnancy (also known as obstetric cholestasis) occurs in around 1% of pregnancies and is generally seen in the third trimester. It is the most common liver disease of pregnancy and is associated with an increased risk of premature birth.

Features:

- pruritus which may be intense and it is typically worse in the palms, soles and abdomen.
- no rash (although skin changes may be seen due to scratching)
- raised bilirubin is seen in > 90% of cases
- clinically detectable jaundice occurs in around 20% of patients

Management:

- ursodeoxycholic acid is used for symptomatic relief
- weekly liver function tests
- women are typically induced at 37 weeks
- vitamin K supplementation

Complications include an increased rate of stillbirth. It is not generally associated with increased maternal morbidity
Recurrence of intrahepatic cholestasis of pregnancy is 45-90% in subsequent pregnancies.

TOPIC: Acute fatty liver of pregnancy

Acute fatty liver of pregnancy is rare complication which may occur in the third trimester or the period immediately following delivery.

Features:

- abdominal pain
- nausea & vomiting
- headache
- jaundice
- hypoglycaemia
- severe disease may result in pre-eclampsia

Investigations:

- ALT is typically elevated e.g. 500 u/l

Management:

- support care
- once stabilised delivery is the definitive management

Gilbert's, Dubin-Johnson syndrome, may be exacerbated during pregnancy

HELLP:

- Haemolysis, Elevated Liver enzymes, Low Platelets

TOPIC: Pregnancy: obesity

Obese women, and their unborn children, are at an increased risk of a number of complications during pregnancy and labour. Obesity is usually defined as a body mass index (BMI) >= 30 kg/m² at the first antenatal visit.

Maternal risks:

- miscarriage
- venous thromboembolism
- gestational diabetes
- pre-eclampsia
- dysfunctional labour, induced labour
- postpartum haemorrhage
- wound infections

There is also a higher caesarean section rate.

Fetal risks:

- congenital anomaly
- prematurity
- macrosomia
- stillbirth
- increased risk of developing obesity and metabolic disorders in childhood
- neonatal death

With respect to weight loss, NICE state:
Explain to women with a BMI of 30 or more at the booking appointment how this poses a risk, both to their health and the health of the unborn child. Explain that they should not try to reduce this risk by dieting while pregnant and that the risk will be managed by the health professionals caring for them during their pregnancy

Management:

- obese women should take 5mg of folic acid, rather than 400mcg
- all obese women should be screened for gestational diabetes with an oral glucose tolerance test (OGTT) at 24-28 weeks
- if the BMI >= 35 kg/m² women should give birth in a consultant-led obstetric unit
- if the BMI >= 40 kg/m² should have an antenatal consultation with an obstetric anaesthetist and a plan made

TOPIC: Alpha feto-protein

Alpha-fetoprotein (AFP) is a protein produced by the developing fetus

Increased AFP	Decreased AFP
Neural tube defects (meningocele, myelomeningocele and anencephaly) Abdominal wall defects (omphalocele and gastroschisis) Multiple pregnancy	Down's syndrome Trisomy 18 Maternal diabetes mellitus

TOPIC: Folic acid

Folic acid is converted to tetrahydrofolate (THF). Green, leafy vegetables are a good source of folic acid.

Functions:

- THF plays a key role in the transfer of 1-carbon units (e.g. methyl, methylene, and formyl groups) to the essential substrates involved in the synthesis of DNA & RNA

Causes of folic acid deficiency:

- Phenytoin
- Methotrexate
- Pregnancy
- Alcohol excess

Consequences of folic acid deficiency:

- Macrocytic, megaloblastic anaemia
- Neural tube defects

Prevention of neural tube defects (NTD) during pregnancy:

- all women should take 400mcg of folic acid until the 12th week of pregnancy
- women at higher risk of conceiving a child with a NTD should take 5mg of folic acid from before conception until the 12th week of pregnancy
- women are considered higher risk if any of the following apply:
    - either partner has a NTD, they have had a previous pregnancy affected by a NTD, or they have a family history of a NTD
    - the woman is taking antiepileptic drugs or has coeliac disease, diabetes, or thalassaemia trait.
    - the woman is obese (defined as a body mass index [BMI] of 30 kg/m^2 or more).

TOPIC: Galactocele

Galactocele typically occurs in women who have recently stopped breastfeeding and is due to occlusion of a lactiferous duct. A build up, of milk creates a cystic lesion in the breast. The lesion can be differentiated from an abscess by the fact that a galactocele is usually painless, with no local or systemic signs of infection.

TOPIC: Reduced fetal-movements

Introduction :
Reduced fetal movements can represent fetal distress, as a method of fetal compensation to reduce oxygen consumption as a response to chronic hypoxia in utero. This is concerning, as it reflects risk of stillbirth and fetal growth restriction. It is believed that there may also be a link between reduced fetal movements and placental insufficiency.

Physiology:
The first onset of recognised fetal movements is known as quickening. This usually occurs between 18-20 weeks-gestation, and increase until 32 weeks-gestation at which point the frequency of movement tends to plateau. Multiparous women will usually experience fetal movements sooner, from 16-18 weeks-gestation. Towards the end of pregnancy, fetal movements should not reduce.

Expectant mothers will usually quickly recognise a pattern to these movements. The nature of the movements themselves can be very variable. There is no established definition for what constitutes reduced fetal movements (RFM), but the RCOG considers less than 10 movements within 2 hours (in pregnancies past 28 weeks-gestation) an indication for further assessment.

Epidemiology :
Fetal movements should be established by 24 weeks gestation.

Reduced fetal movements is a fairly common presentation, affecting up to 15% of pregnancies. 3-5% of pregnant women will have recurrent presentations with RFM.

Risk factors for reduced fetal movements:

- Posture
  - There can be positional changes in fetal movement awareness, generally being more prominent during lying down and less when sitting and standing
- Distraction
  - Awareness of fetal movements can be distractable, and if a woman is busy or concentrating on something else, these can be less prominent
- Placental position
  - Patient with anterior placentas prior to 28 weeks-gestation may have lesser awareness of fetal movements
- Medication
  - Both alcohol and sedative medications like opiates or benzodiazepines can temporarily cause reduced fetal movements
- Fetal position
  - Anterior fetal position means movements are less noticeable
- Body habitus
  - Obese patients are less likely to feel prominent fetal movements
- Amniotic fluid volume
  - Both oligohydramnios and polyhydramnios can cause reduction in fetal movements
- Fetal size
  - Up to 29% of women presenting with RFM have a SGA fetus

Investigations :
Fetal movements are usually based solely on maternal perception, though it can also be objectively assessed using handheld Doppler or ultrasonography.

As per RCOG Green-top guidelines, investigations are dependent of gestation at onset of RFM.

- If past 28 weeks-gestation:
  - Initially, handheld Doppler should be used to confirm fetal heartbeat.
  - If no fetal heartbeat detectable, immediate ultrasound should be offered.
  - If fetal heartbeat present, CTG should be used for at least 20 minutes to monitor fetal heart rate which can assist in excluding fetal compromise.
  - If concern remains, despite normal CTG, urgent (within 24 hours) ultrasound can be used. Ultrasound assessment should include abdominal circumference or estimated fetal weight (to exclude SGA), and amniotic fluid volume measurement
- If between 24 and 28 weeks-gestation, a handheld Doppler should be used to confirm presence of fetal heartbeat.
- If below 24 weeks gestation, and fetal movements have previously been felt, a handheld Doppler should be used.
- If fetal movements have not yet been felt by 24 weeks, onward referral should be made to a maternal fetal medicine unit.

If RFM are recurrent, further investigations are also required to consider structural or genetic fetal abnormalities.

Prognosis:
Concern regarding absent or reduced fetal movements stems for the potential for this presentation to represent fetal distress or impending demise. Between 40-55% of women who suffer from stillbirth experience reduced fetal movements prior to diagnosis.

However, in 70% of pregnancies with a single episode of reduced fetal movement, there is no onward complication.

TOPIC: Rhesus negative pregnancy

A basic understanding of the pathophysiology is essential to understand the management of Rhesus negative pregnancies

- along with the ABO system the Rhesus system is the most important antigen found on red blood cells. The D antigen is the most important antigen of the rhesus system
- around 15% of mothers are rhesus negative (Rh -ve)
- if a Rh -ve mother delivers a Rh +ve child a leak of fetal red blood cells may occur
- this causes anti-D IgG antibodies to form in mother
- in later pregnancies these can cross placenta and cause haemolysis in fetus
- this can also occur in the first pregnancy due to leaks

Prevention:

- test for D antibodies in all Rh -ve mothers at booking
- NICE (2008) advise giving anti-D to non-sensitised Rh -ve mothers at 28 and 34 weeks
- the evidence base suggests that there is little difference in the efficacy of single-dose (at 28 weeks) and double-dose regimes (at 28 & 34 weeks). For this reason the RCOG in 2011 advised that either regime could be used 'depending on local factors'
- anti-D is prophylaxis - once sensitization has occurred it is irreversible
- if event is in 2nd/3rd trimester give large dose of anti-D and perform Kleihauer test - determines proportion of fetal RBCs present.

Anti-D immunoglobulin should be given as soon as possible (but always within 72 hours) in the following situations:

- delivery of a Rh +ve infant, whether live or stillborn
- any termination of pregnancy
- miscarriage if gestation is > 12 weeks
- ectopic pregnancy (if managed surgically, if managed medically with methotrexate anti-D is not required)
- external cephalic version
- antepartum haemorrhage
- amniocentesis, chorionic villus sampling, fetal blood sampling
- abdominal trauma

Tests:

- all babies born to Rh -ve mother should have cord blood taken at delivery for FBC, blood group & direct Coombs test
- Coombs test: direct antiglobulin, will demonstrate antibodies on RBCs of baby
- Kleihauer test: add acid to maternal blood, fetal cells are resistant

Affected fetus:

- oedematous (hydrops fetalis, as liver devoted to RBC production albumin falls)
- jaundice, anaemia, hepatosplenomegaly
- heart failure, kernicterus.
- treatment: transfusions, UV phototherapy

TOPIC: Ultrasound in pregnancy

A nuchal scan is performed at 11-13 weeks. Causes of an increased nuchal translucency include:

- Down's syndrome
- congenital heart defects
- abdominal wall defects

Causes of hyperechogenic bowel:

- cystic fibrosis
- Down's syndrome
- cytomegalovirus infection

Symphysis-fundal height

- The symphysis-fundal height (SFH) is measured from the top of the pubic bone to the top of the uterus in centimetres
- It should match the gestational age in weeks to within 2 cm after 20 weeks, e.g. if 24 weeks then the a normal SFH = 22 to 26 cm

TOPIC: Down's syndrome: antenatal testing

NICE issued updated guidelines on antenatal care in 2021 including advice on screening for Down's syndrome

- **The combined test is now standard**
  - these tests should be done between 11 - 13+6 weeks
  - nuchal translucency measurement + serum B-HCG + pregnancy-associated plasma protein A (PAPP-A)
  - Down's syndrome is suggested by ↑ HCG, ↓ PAPP-A, thickened nuchal translucency
  - trisomy 18 (Edward syndrome) and 13 (Patau syndrome) give similar results but the hCG tends to lower
- **Quadruple test**
- If women book later in pregnancy the quadruple test should be offered between 15 - 20 weeks
  - quadruple test: alpha-fetoprotein, unconjugated oestriol, human chorionic gonadotrophin and inhibin A
- Once a screening test has been performed, the chance of the fetus having Down's syndrome is calculated using software that takes into account maternal factors such as age, weight and family origin, along with gestation of the pregnancy.

Results of combined or quadruple tests

Both the combined and quadruple tests return either a 'lower chance' or 'higher chance' result

- 'lower-chance': 1 in 150 chance or more e.g. 1 in 300
- 'higher-chance': 1 in 150 chance or less e.g. 1 in 100
- If a patient has a lower risk result they will not be offered a further test for Down syndrome.
- If the patient has a higher risk result they will be offered a diagnostic test to find out whether their baby has Down syndrome or not. It is the patient's choice whether or not to have the further test.

Interpreting the results of quadrapule tests

	Alpha-fetoprotein	Unconjugated oestriol	Human chorionic gonadotrophin	Inhibin A
Down's syndrome	↓	↓	↑	↑
Edward's syndrome	↓	↓	↓	↔
Neural tube defects	↑	↔	↔	↔

Non-invasive prenatal screening test (NIPT)

If a woman has a 'higher chance' results she will be offered a second screening test (NIPT) or a diagnostic test (e.g. amniocentesis or chorionic villus sampling (CVS). Given the non-invasive nature of NIPT and extremely high sensitivity and specificity, it is likely this will be the preferred choice for the vast majority of women.

NIPT:

- analyses small DNA fragments that circulate in the blood of a pregnant woman (cell free fetal DNA, cffDNA)
- cffDNA derives from placental cells and is usually identical to fetal DNA

- analysis of cffDNA allows for the early detection of certain chromosomal abnormalities
- sensitivity and specificity are very high for trisomy 21 (>99%) and similarly high for other chromosomal abnormalities
- private companies (e.g. Harmony) offer NIPT screening from 10 weeks gestation

Summary of important points

SCREEING TESTS

- **The combined test** offers the most effective and safe method of screening for women who attend in the first trimester.
- **The quadruple test** is the best test for women who first present in the second trimester.
- NIPT

DIAGNOSTIC

- Diagnostic testing using either chorionic villus sampling (if less than 13 weeks of gestation) or amniocentesis (if beyond 15 weeks of gestation). These procedures carry a risk of miscarriage (0.5-1% excess miscarriage risk for amniocentesis; 1-2% for chorionic villus sampling).

TOPIC: GROUP B STREPTICOCCUS

Group B *Streptococcus* (GBS) is the most common cause of early-onset severe infection in the neonatal period. It is thought around 20-40% of mothers have GBS present in their bowel flora and may therefore be thought of as 'carriers' of GBS. Infants may be exposed to maternal GBS during labor and subsequently develop potentially serious infections.

Risk factors for Group B *Streptococcus* (GBS) infection:

- prematurity
- prolonged rupture of the membranes
- previous sibling GBS infection
- maternal pyrexia e.g. secondary to chorioamnionitis

Screening

- universal screening for GBS should not be offered to all women (RCOG); in contrast, UpToDate recommends universal screening.
- the guidelines also state a maternal request is not an indication for screening
- women who've had GBS detected in a previous pregnancy should be informed that their risk of maternal GBS carriage in this pregnancy is 50%. They should be offered intrapartum antibiotic prophylaxis (IAP) or testing in late pregnancy and then antibiotics if still positive
- if women bs for GBS this should be offered at 35-37 weeks or 3-5 weeks prior to the anticipated delivery date are to have swab

Management

The Royal College of Obstetricians and Gynaecologists (RCOG) published guidelines on GBS in 2017.The main points are as follows:

- IAP should be offered to women with a previous baby with early- or late-onset GBS disease
- IAP should be offered to women in preterm labor regardless of their GBS status
- women with a pyrexia during labor (>38ºC) should also be given IAP
- benzylpenicillin is the antibiotic of choice for GBS prophylaxis

TOPIC: TOXOPLASMOSIS

Toxoplasmosis is an acquired protozoan parasite infection by toxoplasma gondii via various settings during childhood and adolescence.

Maternal Clinical Features:

- Asymptomatic
- Flu-like symptoms: malaise, fever, chills, sweats, headaches, and pharyngitis
- Hepatosplenomegaly
- Diffuse non-pruritis maculopapular rash

Congenital Toxoplasmosis Clinical Features:

Asymptomatic
Moderate symptoms: intracranial calcifications – peripheral retinochoroiditis
Severe symptoms: disseminated form – hydrocephalus – macular retinochoroiditis

Transmission to Fetus:

- Vertical (transplacental) transmission of tachyzoites to fetus after acute maternal infection in pregnancy or after reactivation of latent infection, less likely, in immunocompromised women; this will lead to congenital infection.

Risk Factors for Vertical Transmission:

- High parasite load
- Maternal infection at advanced gestational age
- Maternal immunocompromised
- High virulence strain

Screening:

- Not indicated routinely for pregnant women.
- Screening is performed only with clinical suspicion of acute infection during pregnancy.
- IgM: if positive or equivocal to confirm by PCR for t. gondii DNA in amniotic fluid

Management:

- Less than 14 weeks: Spiramycin
- At/more than 14 weeks: Pyrimethamine-sulfadiazine

Breast Feeding:

- No contraindications

TOPIC: Breastfeeding

Advantages	Disadvantages
Mother • bonding • involution of uterus • protection against breast and ovarian cancer • cheap, no need to sterilise bottle • contraceptive effect (unreliable) Immunological • IgA (protects mucosal surfaces), lysozyme (bacteriolytic enzyme) and lactoferrin (ensures rapid absorption of iron so not available to bacteria) • reduced incidence of ear, chest and gastro-intestinal infections • reduced incidence of eczema and asthma • reduced incidence of type 1 diabetes mellitus Reduced incidence of sudden infant death syndrome Baby is in control of how much milk it takes	Transmission of drugs Transmission of infection (e.g. HIV) Nutrient inadequacies (prolonged breast feeding may lead to vitamin D deficiency) Vitamin K deficiency Breast milk jaundice

TOPIC: Breastfeeding problems

- The World Health Organization (WHO) recommends exclusive breastfeeding until an infant is 6 months of age, with the introduction of solid food around this time and continued breastfeeding up to 2 years of age or longer.
    - Benefits to the infant include nutritional benefits, a reduction in the incidence and severity of infections, asthma, and atopic eczema.
    - Benefits to the mother include reduced rates of breast and ovarian cancer, diabetes mellitus, hypertensive heart disease and reduced incidence of obesity.
- Breastfeeding problems which may lead to a mother stopping breastfeeding include:
    - Breast pain
    - Nipple pain
    - Low milk supply (true and perceived).
    - Oversupply of milk
- Effective and efficient breastfeeding requires optimal infant positioning and attachment to the breast. If this is not achieved, there may be:
    - Reduced milk transfer.
    - Reduced stimulation of breast milk production.
    - Insufficient milk to match the infant's needs, leading to possible dehydration and faltering growth.
    - Nipple pain and damage leading to other possible breastfeeding problems.

Assessment of breastfeeding problems?

Assessment should include history and examination of both the mother and infant.

- **Ask about:**
    - **Breastfeeding history, including:**
        - Concerns the parents have about their baby's feeding.
        - Pattern of breastfeeding (frequency, duration, night feeds, one or both breasts offered).
        - Presence of breast or nipple sensitivity before pregnancy.
        - Milk supply issues (engorgement, low or oversupply).
        - Expressing of milk (frequency, hand or pump used).
        - Other fluids or foods given (when started, quantity, and frequency).
        - Use of nipple shields or breast shells.
        - Maternal beliefs, concerns, and expectations about breastfeeding, including sources of support and barriers to breastfeeding.
        - Previous breastfeeding experiences, problems, and pain.
    - **Breast and/or nipple pain history, including:**
        - Time of onset post-partum.
        - Presence of nipple trauma (abrasions, fissures, bleeding).
        - Timing (for example with attachment, during breastfeeds, between breastfeeds, with expressing of milk) and whether intermittent or constant.
        - Location (for example unilateral or bilateral, nipple and/or breast, superficial or deep).
        - Character and severity (for example burning, itching, sharp, dull).
        - Presence of associated signs and symptoms (such as fever, breast skin changes, malaise, nipple colour changes, nipple shape or appearance after feeds).
        - Exacerbating or relieving factors (such as cold, heat, massage or touch).
        - Previous treatments (such as analgesia, topical or oral drugs including antibiotics).

- **Maternal history, including:**
  - Complications during pregnancy, labour, or post-partum.
  - Known medical conditions, such as thyroid disorders, diabetes, Raynaud's phenomenon, eczema or psoriasis, recent Candida or bacterial infection, chronic pain syndromes, and family history of ankyloglossia (tongue-tie).
  - Previous breast surgery.
  - Medications and allergies.
  - Alcohol and smoking.
  - Associated insomnia, stress, anxiety, or depression.
- **Infant history, including:**
  - Birth gestation, birth trauma and known medical conditions or congenital abnormalities.
  - Birth weight, weight gain, general health and behaviour (for example settled or crying).
  - Number of wet and dirty nappies (including stool colour).
  - Behaviour at the breast (for example rhythmic sucking and audible swallowing; pulling, biting, coughing, breathlessness, or sleepiness).
  - Gastrointestinal symptoms which may suggest oversupply of milk, gastro-oesophageal reflux disease, or cows' milk protein allergy.
  - Known ankyloglossia or cleft lip and/or palate and any treatment received.
  - Use of a dummy or pacifier.

- **Examine the mother:**
  - Carry out a general examination, look for signs of infection or systemic illness.
  - Inspect the nipples and breasts to identify flat or inverted nipples, skin damage, discharge, crusting, fissuring, discolouration, peau d'orange, individual lesions or rashes.
  - Palpate the breasts to identify any breast lumps, firmness, or tenderness.
- **Examine the infant:**
  - Carry out a general examination, look for signs of dehydration, jaundice, lethargy, and inconsolability.
  - Check head and neck symmetry, facial features and oral anatomy (ankyloglossia, palate, jaw, and lips), muscle tone, neurological maturity, and behaviour.
  - Look for signs of oral Candida infection.
- **Ensure that a person with appropriate training and expertise (such as a health visitor or breastfeeding specialist) observes the woman:**
  - **Breastfeeding to check:**
    - Maternal and infant positioning.
    - Attachment (mouth wide-open with lips everted).
    - Behaviour at the breast.
    - Pattern of sucking.
    - Shape and colour of nipples after feeding.
  - **Expressing milk to check**:
    - Hand and/or breast pump expressing technique.
    - Breast shield/breast pump flange fit — nipple or breast trauma may occur due to breast pump use (for example from improper flange fit, excessive high- pressure suction, or prolonged duration of use).
- **Consider arranging a skin swab of the nipple and/or breast for microscopy and culture, if there is:**
  - Persistent nipple cracking or fissures.
  - Suspected bacterial or candida infection.
  - Breast or nipple pain disproportionate to examination findings.
  - An uncertain diagnosis — discuss with a specialist if unsure. In some cases culture of breast milk may be indicated.

How to diagnose the cause of breast pain?

- **Engorgement**
  - Breast pain typically starts in the first few days after birth, is often bilateral and worse before a feed.
  - Infant attachment may be difficult due to breast fullness and milk flow may be reduced.
  - The whole breast is typically swollen and oedematous and may be shiny, diffusely erythematous and may leak excessively.
  - The nipple may be stretched and look flat in appearance.
  - The infant may cough and pull off the breast on feeding or clamp down on the nipple during feeding to control flow.
- **Blocked ducts**
  - May present with a localized, tender cord of tissue in one breast (usually a few centimetres in diameter), which may be relieved by expression of milk — overlying skin may be erythematous.
  - A small white spot (bleb) about 1 mm in diameter may be present at the end of the nipple.
- **Galactocele**
  - A galactocele may result from a blocked duct, and typically presents as a smooth, round, painless breast swelling which causes milky nipple discharge when pressed.
- **Ductal infection**
  - Ductal infection may present with deep burning, aching, or shooting breast pain that occurs during and between feeds. There may be erythematous, flaky or shiny skin of the nipple, or a nipple fissure.
  - Purulent exudate or crust may suggest associated bacterial infection.
- **Mastitis and breast abscess**
  - The woman often has a fever and is systemically unwell — mastitis may present with a hard painful swelling in a wedge-shaped distribution in one breast, with erythema of the overlying skin.
  - A breast abscess may present with a worsening painful breast lump, which may be fluctuant, and the overlying skin is often erythematous and warm. There may be a persistent fever and systemic symptoms.

How to diagnose the cause of nipple pain?

- **Physiological milk let-down pain:**
  - There is often pain for the first few minutes of breastfeeding, which resolves with continuation of the feed.
  - Pain often improves during the first few weeks of breastfeeding.
- **Nipple damage:**
  - Suboptimal positioning and attachment typically, causes nipple pain at the start of a breastfeed which continues throughout the feed.
  - Pressure from suckling may cause blanching and compression of the nipple, and fissuring across the top of the nipple or around the base. There may be flattening of the nipple from side to side, with a pressure line across the tip.
  - Nipple damage may also occur due to infant ankyloglossia (tongue tie) or breast pump misuse or trauma.
- **Nipple infection:**
  - Bacterial infection may present with purulent nipple discharge, crusting, redness, and fissuring and is often associated with skin trauma such as persistent cracks and fissures.

    - Candida infection may cause bilateral burning nipple pain, itching, and hypersensitivity of the nipple, especially during and soon after feeds.
        - The nipple may be red, shiny, swollen, or fissured.
        - Typically, the pain does not resolve despite improved positioning and attachment, or follows a period of pain-free breastfeeding.
        - Clinical signs of candida infection may also be present in the breast-fed infants mouth.
    - Viral infection (such as herpes simplex, herpes zoster or varicella zoster) may present with vesicular rash affecting the breast.
- **Skin conditions:**
    - Eczema, psoriasis and contact dermatitis of the areola and nipple may cause itching of the skin.
        - Eczema typically causes a bilateral red, dry, scaly rash which may have lichenified (thickened) areas, which tend to spare the base of the nipple.
        - Psoriasis typically causes red plaques with clearly demarcated borders, which may have a fine overlying scale.
        - Irritant dermatitis may occur due to soaps, nipple creams and other substances in direct contact with the breast.
    - Paget's disease of the nipple may mimic eczema, but is usually unilateral, persistent, and unresponsive to treatment for eczema. In addition, there may be skin ulceration or erosions.
- **Nipple vasospasm or Raynaud's disease of the nipple:**
    - Nipple pain (may be described as shooting, throbbing or burning) is typically intermittent, and present during and immediately after breastfeeds, and in between feeds if exposed to cold temperatures.
    - Blanching of the nipple may be followed by cyanosis and/or erythema.
    - Nipple pain resolves when the nipple returns to its normal colour.
    - There may be a personal or family history of Raynaud's phenomenon of the digits.

How to diagnose the cause of low milk supply?

- **True maternal low milk supply is unusual, and there may be a subjective maternal perception of insufficient milk supply, if other causes have been excluded or are unlikely.**
- **Insufficient access to the breast:**
    - Suggested by short or infrequent feeds, and/or no night feeds.
    - Use of a dummy; or giving supplementary feeds other than breast milk may also contribute.
    - Maternal depression, stress, and/or anxiety may result in a reduced response to infant feeding cues and a reduced frequency of feeds, which leads to reduced stimulation of milk production.
- **Suboptimal infant positioning and attachment:**
    - Suggested by nipple pain/trauma; frequent feeding more than every 2 hours; no long intervals between feeds; feeding for less than five minutes or longer than 40 minutes duration.
    - The infant may be generally unsettled, have faltering growth, or show signs of dehydration.
- **Maternal prolactin deficiency:**
    - Causes of true low milk supply due to maternal prolactin deficiency may include drugs, thyroid disorders, retained placenta, alcohol use, and eating disorders.
- **Maternal anatomical conditions** such as hypoplastic breasts causing a lack of glandular tissue, or a history of breast surgery.

How to diagnose the cause of milk oversupply?

- **Suboptimal infant positioning and attachment**
  - The infant may not remove milk efficiently so suckles a lot, stimulating the breast to produce excessive milk.
- **Breastfeeding pattern**
  - Moving the infant too early to the second breast before they have finished feeding from the first breast.
  - Overstimulation due to excessive expression (by hand or pump) between breastfeeds.
- **The mother may describe:**
  - Breast fullness and possible engorgement or blocked ducts.
  - A painful, forceful milk let-down reflex.
  - Milk leakage and/or milk spraying from the opposite breast when feeding.
- **The infant may:**
  - Choke and splutter when let-down occur.
  - Clamp down on the nipple or pull off the breast during feeds.
  - Have colic, or frequent explosive loose stools.
  - Faltering growth if unable to feed adequately.

Management of breastfeeding problems

- **For all women with breastfeeding problems:**
  - Ensure that a person with appropriate training and expertise (such as a health visitor or breastfeeding specialist) observes the woman breastfeeding and expressing milk to check and give advice on:
    - Optimal infant positioning and attachment to the breast.
    - Hand and/or breast pump milk expressing technique; level of suction; and fit of the flange.
    - Initial breast fullness and nipple discomfort in the first few days post-partum which may be normal as the milk 'comes in' and should improve as maternal milk supply adjusts to the infant's needs.
    - Responsive feeding.
- **Offer written information on optimal infant positioning and attachment**
- **Following assessment and advice on optimal infant positioning and attachment to the breast, if there is suspected:**
  - **Engorgement, advise the woman on:**
    - Feeding the infant with no restrictions on the frequency or length of feeds.
    - Self-management techniques such as simple analgesia (for example paracetamol) for pain relief, breast massage after feeds, and minimal expressing of milk to relieve full breasts (excessive expression of milk may induce oversupply). Hand expressing a little milk to soften the areola prior to feeding may help if the baby is struggling to latch on due to breast engorgement.
    - The use of heat packs or a warm shower before feeding or expressing milk which can stimulate milk let-down, or the use of cold packs after feeding or expressing, to relieve pain and oedema.
    - Wearing a well-fitting bra and clothing that does not restrict the breasts.
  - **Blocked ducts:**
    - Advise on feeding from the affected breast frequently; the use of heat packs or a warm shower for symptom relief; wearing a well-fitting bra and clothing that does not restrict the breasts; and gentle massage of the breast (using a

firm movement towards the nipple) while the baby is feeding, to help relieve the obstruction.
    - If there is a white spot (bleb) on the nipple tip, advise on bathing then rubbing the area with a warm, damp towel.
    - Advise the woman on signs of mastitis (which can develop as a complication of blocked ducts) and when to seek medical advice
    - For a persistent breast mass, consider the possibility of an alternative diagnosis (such as breast cancer) and the need for referral.
- **Galactocele:**
    - Advise the woman to continue breastfeeding.
    - Discuss with a breast specialist investigation to confirm diagnosis and treatment options if needed.
- **Ductal infection:**
    - Ensure that other causes of persistent breast/nipple pain have been excluded.
    - Seek specialist advice regarding the need for investigations (such as nipple and breast milk cultures) and antimicrobial treatment — depending on the specific clinical situation oral antibiotic or oral antifungal treatment may be indicated.
- **Nipple damage:**
    - Suboptimal infant positioning and attachment is the most common cause of nipple damage — ensure the woman has been reviewed by a person with appropriate training and expertise (such as a health visitor or breastfeeding specialist) and given advice and support.
    - Arrange paediatric referral if there is an infant anatomical abnormality such as ankyloglossia (tongue-tie), for consideration of division of the lingual frenulum.
    - Advise on stopping the use of nipple shields or breast shells, if appropriate, as these may contribute to incorrect positioning and attachment.
    - Advise on considering the application of expressed breast milk, if the nipple skin is cracked or fissured.
- **Nipple infection**
    - For mild localised bacterial infection, consider a topical antibiotic such as fusidic acid (or mupirocin if antimicrobial resistance is suspected). Breastfeeding from the affected nipple should be avoided during treatment with fusidic acid and expression of milk to avoid engorgement discussed. If mupirocin is used the area should be washed thoroughly before breastfeeding.
    - For Candida infection, provide treatment for the woman (such as miconazole 2% cream) and the infant at the same time to prevent re-infection.
    - If the nipples are red and inflamed, consider prescribing hydrocortisone 1% cream in addition to topical antimicrobials.
    - If a viral infection such as *Herpes simplex* or *Varicella zoster* is suspected discuss management with a specialist – be aware that transmission to an infant can cause serious illness, especially in neonates or those with immunocompromise.
- **Skin conditions:**
    - If there is suspected eczema, advise on avoiding possible triggers if appropriate, such as lanolin, beeswax, or chamomile in nipple creams; soap, shampoo, and fragrances. Advise on the use of regular emollients and application of a topical corticosteroid to the nipples, immediately after a breastfeed, the potency depending on the severity of the eczema.

        - If there is suspected psoriasis, advise on the use of regular emollients and topical treatment to the nipples, immediately after a breastfeed.
        - If there is suspected Paget's disease of the nipple, arrange urgent referral to a breast specialist using a suspected cancer pathway.
    - **Nipple vasospasm or Raynaud's disease of the nipple:**
        - Advise on avoiding exposure to the cold, wearing warm clothing, and breastfeeding in a warm environment and self-management techniques such as the use of heat packs or a warm shower following a breastfeed or when there is breast pain.
        - Advise on avoiding caffeine and stopping smoking which can cause vasoconstriction, if appropriate.
        - If symptoms persist, consider discussion with/referral to an appropriate specialist, for consideration of a trial of oral nifedipine (off-label indication).
    - **Low milk supply:**
        - Arrange a paediatric referral (with urgency depending on the clinical situation) for the infant if there are concerns about dehydration, faltering growth, or infant development.
        - Manage any underlying cause in the infant or woman that may be contributing to a low milk supply.
        - Ensure that an assessment by a skilled lactation specialist has taken place to provide information on optimal positioning and attachment, skin-to-skin contact, feeding frequency, effective breast drainage and expressing breastmilk.
        - If these measures do not improve a true low milk supply and maternal prolactin deficiency is a possible cause, discuss with/refer to an endocrinologist for further investigations/management.
    - **Milk oversupply:**
        - Ensure that the woman is aware of early feeding cues that suggest an infant is hungry, so that breastfeeds are initiated at appropriate times.
        - Advise that if the infant is unable to attach effectively to the breast due to breast fullness, it may be helpful to express a small amount of milk (for example by hand expression) until the flow slows down, and then try to attach the infant to the breast. If a forceful let-down reflex continues, advise the woman that leaning back to breastfeed may be helpful.
        - Ensure the woman has been assessed by person with appropriate training and expertise (such as a health visitor or breastfeeding specialist) to ensure optimal positioning and attachment. Feeding from one breast for each feed may be advised to help reduce milk supply.
        - Advise on avoiding milk expression by hand or breast pump between feeds, if possible, to reduce overstimulation of milk supply.
- **If symptoms do not improve as expected with appropriate treatment, consider discussion with/referral to a breast specialist for further advice on management.**

TOPIC: Breastfeeding: contraindications

The major breastfeeding contraindications tested in exams relate to drugs (see below). Other contraindications of note include:

- galactosaemia
- viral infections - this is controversial with respect to HIV in the developing world. This is because there is such an increased infant mortality and morbidity associated with bottle feeding that some doctors think the benefits outweigh the risk of HIV transmission

Drug contraindications

The following drugs can be given to mothers who are breastfeeding:

- antibiotics: penicillins, cephalosporins, trimethoprim
- endocrine: glucocorticoids (avoid high doses), levothyroxine*
- epilepsy: sodium valproate, carbamazepine
- asthma: salbutamol, theophyllines
- psychiatric drugs: tricyclic antidepressants, antipsychotics**
- hypertension: beta-blockers, hydralazine
- anticoagulants: warfarin, heparin
- digoxin

The following drugs should be avoided:

- antibiotics: ciprofloxacin, tetracycline, chloramphenicol, sulphonamides
- psychiatric drugs: lithium, benzodiazepines
- aspirin
- carbimazole
- methotrexate
- sulfonylureas
- cytotoxic drugs
- amiodarone

*The BNF advises that the amount is too small to affect neonatal hypothyroidism screening

**clozapine should be avoided

TOPIC: Breech presentation

In a breech presentation the caudal end of the fetus occupies the lower segment. Whilst around 25% of pregnancies at 28 weeks are breech it only occurs in 3% of babies near term. A frank breech is the most common presentation with the hips flexed and knees fully extended. A footling breech, where one or both feet come first with the bottom at a higher position, is rare but carries a higher perinatal morbidity

Risk factors for breech presentation:

- uterine malformations, fibroids
- placenta praevia
- polyhydramnios or oligohydramnios
- fetal abnormality (e.g. CNS malformation, chromosomal disorders)
- prematurity (due to increased incidence earlier in gestation)

Cord prolapse is more common in breech presentations.

Management:

- if < 36 weeks: many fetuses will turn spontaneously
- if still breech at 36 weeks NICE recommend external cephalic version (ECV)- this has a success rate of around 60%. The RCOG recommend ECV should be offered from 36 weeks in nulliparous women and from 37 weeks in multiparous women
- if the baby is still breech then delivery options include planned caesarean section or vaginal delivery

Information to help decision making - the RCOG recommend:

- 'Women should be informed that planned caesarean section carries a reduced perinatal mortality and early neonatal morbidity for babies with a breech presentation at term compared with planned vaginal birth.'
- 'Women should be informed that there is no evidence that the long-term health of babies with a breech presentation delivered at term is influenced by how the baby is born.'

RCOG absolute contraindications to ECV:

- where caesarean delivery is required
- antepartum haemorrhage within the last 7 days
- abnormal cardiotocography
- major uterine anomaly
- ruptured membranes
- multiple pregnancy

TOPIC: Parity and gravity

Parity is the number of pregnancies a woman has had which have been carried to a viable age; in the UK this is 24 weeks. The number after '+' is the number of pregnancies which have not been carried to a viable age. It can be thought of the number of fetuses/babies which have come from her, in contrast to gravida which is the number of times the uterus has contained a foetus. For example, twins are counted as Gravida 1 Parity 2. Parity also does not increase until the foetus is born but gravida technically increases from conception (though in practice from a woman's first appointments with her doctor).

TOPIC: Postpartum mental health problems

Postpartum mental health problems range from the 'baby-blues' to puerperal psychosis.

The **Edinburgh Postnatal Depression Scale** may be used to screen for depression:

- 10-item questionnaire, with a maximum score of 30
- indicates how the mother has felt over the previous week
- score > 13 indicates a 'depressive illness of varying severity'
- sensitivity and specificity > 90%
- includes a question about self-harm

'Baby-blues'	Postnatal depression	Puerperal psychosis
Seen in around 60-70% of women Typically seen 3-7 days following birth and is more common in primips Mothers are characteristically anxious, tearful and irritable	Affects around 10% of women Most cases start within a month and typically peaks at 3 months Features are similar to depression seen in other circumstances	Affects approximately 0.2% of women Onset usually within the first 2-3 weeks following birth Features include severe swings in mood (similar to bipolar disorder) and disordered perception (e.g. auditory hallucinations)
Reassurance and support, the health visitor has a key role	As with the baby blues reassurance and support are important NICE CKS state *'Most women with the baby blues will not require specific treatment other than reassurance'* Cognitive behavioural therapy may be beneficial. Certain SSRIs such as sertraline and paroxetine* may be used if symptoms are severe** - whilst they are secreted in breast milk it is not thought to be harmful to the infant	Admission to hospital is usually required, ideally in a Mother & Baby Unit There is around a 25-50% risk of recurrence following future pregnancies

*Paroxetine is recommended by SIGN because of the low milk/plasma ratio
**fluoxetine is best avoided due to a long half-life

TOPIC: Puerperal pyrexia

Puerperal pyrexia may be defined as a temperature of > 38ºC in the first 14 days following delivery.

Causes:

- endometritis: most common cause
- urinary tract infection
- wound infections (perineal tears + caesarean section)
- mastitis
- venous thromboembolism

Management:

- if endometritis is suspected the patient should be referred to hospital for intravenous antibiotics (clindamycin and gentamicin until afebrile for greater than 24 hours)

TOPIC: Postpartum haemorrhage

Postpartum haemorrhage (PPH) is defined as **blood loss of > 500** mlafter a vaginal delivery and may be primary or secondary.

- **Primary postpartum haemorrhage:**
  Primary PPH occurs within 24 hours. It affects around 5-7% of deliveries.
  The causes of PPH are said to be the 4 Ts:

- **T**one (uterine atony): the vast majority of cases
- **T**rauma (e.g. perineal tear)
- **T**issue (retained placenta)
- **T**hrombin (e.g. clotting/bleeding disorder

1. **Risk factors for primary PPH include*:**

- Previous PPH, prolonged labor, pre-eclampsia, polyhydramnios, increased maternal age
- Emergency Caesarean section, macrosomia
- Placenta praevia, placenta accreta
- The effect of parity on the risk of PPH is complicated. It was previously thought multiparity was a risk factor but more modern studies suggest nulliparity is actually a risk factor
  Management:

- PPH is a life-threatening emergency - senior members of staff should be involved immediately
- **ABC approach**
  - Two peripheral cannulae, 14 gauge
  - Lie the woman flat
  - Bloods including group and save
  - Commence warmed crystalloid infusion
- **Mechanical:**
  - Palpate the uterine fundus and rub it to stimulate contractions ('rubbing up the fundus')
  - Catheterisation to prevent bladder distension and monitor urine output
- **Medical:**
  - IV oxytocin: slow IV injection followed by an IV infusion
  - Ergometrine slow IV or IM (unless there is a history of hypertension)
  - Carboprost IM (unless there is a history of asthma)
  - Misoprostol sublingual
  - There is also interest in the role tranexamic acid may play in PPH
- **surgical:** if medical options fail to control the bleeding, then surgical options will need to be urgently considered
  - The RCOG state that the intrauterine balloon tamponade is an appropriate first-line 'surgical' intervention for most women where uterine atony is the only or main cause of haemorrhage
  - Other options include: B-Lynch suture, ligation of the uterine arteries or internal iliac arteries
  - If severe, uncontrolled haemorrhage then a hysterectomy is sometimes performed as a life-saving procedure.

Secondary postpartum haemorrhage:
Secondary PPH occurs between 24 hours - 6 weeks. It is typically due to retained placental tissue or endometritis.

TOPIC: Caesarean delivery

Indications for caesarean section:

- Fetal distress in the first stage of labour
- Failure of progress during the first stage of labour
- Malpresentation - eg, breech, transverse lie.
- All cases of placenta praevia
- Multiple pregnancy: Triplets, conjoint twins, monochromic monoamniotic twins and other cases of twins with non-vertex presentation of the first twin.
- Failed induction of labor.
- Repeat CS: Previous 2 or more CS, previous classical CS and previous LSCS if the patient refused vaginal birth after CS
- Soft tissue obstruction by cervical fibroid or ovarian cyst.
- Maternal infection (eg, herpes, HIV)

Maternal request for caesarean section is NOT on its own an indication; reasons should be explored, discussed and recorded.

- If the request is due to anxiety about childbirth, a referral should be offered to a healthcare professional with expertise in providing perinatal mental health support to help her address her anxiety in a supportive manner.

An individual clinician has the right to decline a request for caesarean section in the absence of an identifiable reason. However, the woman's decision should be respected and she should be offered referral for a second opinion.

TOPIC: Postpartum changes

Involution of uterus in postpartum period:
The uterus increases in size and weight during pregnancy due to the effects of oestrogen, progesterone and chronic stretching on the myometrium due to the growing fetus.

After birth the uterus weighs about 1 kg and the uterine fundus is at the level of the umbilicus. After about two weeks the uterus is no longer palpable abdominally. The weight of the uterus drops to about 50-60 g at about 6 weeks after delivery.

After delivery there is a continuation of uterine activity and for about the first twelve hours the mother may experience regular, strong, well co-ordinated contractions that are similar to those of labour. These contractions gradually decrease in intensity and frequency after this period. Contractions are more painful, in general, in the multigravida. It is not unusual for the mother to experience contractions during breast feeding in early puerperium.

Involution of cervix and vagina post pregnancy:
The cervix is patulous and haemorrhagic immediately after delivery. After about 1 week post-delivery the cervix has contracted to about 1-2 cm dilatation. By the end of the end of the second week there is closure of the cervical os. Cervical lacerations heal spontaneously and the external os becomes a transverse slit.

The vagina returns to almost it's prepartum condition after about 3 weeks.

Perineal injuries (eg tears, episiotomy) cure rapidly and are usually healed after about 1 week.

Involution of placenta site post pregnancy:
The separation of the placenta occurs in the decidual layer. After delivery the placental site contracts to about 50% of its original area. The contraction and retraction of the placental site causes occlusion of the spiral arterioles and so is important in the prevention of postpartum haemorrhage. There is infiltration of the placental bed by inflammatory cells and this superficial layer is shed.

The uterine cavity is re-epithelialized, except for the placental bed, after about 10 days. Re-epithelization of the area of the placental bed does not occur until about 6 weeks post-delivery. During the period of re-epithelization the trophoblast and decidua are shed as a uterine discharge termed lochia. In the first 2-3 days post-delivery the lochia is blood stained. After this initial discharge the lochia becomes serous and pale and lasts for about 20 days. A thicker, more mucoid, discharge then may occur for a further 4-8 weeks.

TOPIC: Diet and fluid in the postpartum period

Postpartum dietry advices:

- Choose a wide variety of foods from all food groups

- During the postpartum period, focus on filling up on healthy sources of:

    - protein
    - fruit
    - vegetables
    - fiber-rich carbs
    - fats like avocados, nuts, and seeds

- Keep in mind that caloric intake and appropriate macronutrient ranges vary depending on your activity levels, body size, and more. If you have a health condition like diabetes, you may need to follow a different dietary pattern in order to optimize blood sugar control. Every woman's nutrition needs are different and depend on many factors

- Stay hydrated all day long

    - It is recommended to drink up to 3 liters of water daily.

- Fueling your body with the right-amount of calories, will help keep your energy and milk supply up.

- According to the CDC trusted source a breastfeeding mother should consume approximately 2,300 to 2,500 calories per day compared to 1,800 to 2,000 calories for a non-breastfeeding woman. However, individual calorie needs are highly variable and depend on body size, age, activity level, and how much you're breastfeeding.

- If you're trying to lose weight while breastfeeding, the Academy of Nutrition and Dietetics says a slow weight loss of 1 pound per week or 4 pounds per month is ideal.

- While the small amount of caffeine that passes from you to the baby through breast milk is not known to adversely affect your infant, the CDC recommends sticking to 300 milligrams or less per day.

- If you're breastfeeding, avoid high mercury seafood and fish like orange roughy, tuna, king mackerel, marlin, shark, swordfish, or tilefish. Instead, opt for salmon, shrimp, cod, tilapia, trout, and halibut, among others.

TOPIC: OLIGOHYDRAMNIOS

- This is too little amniotic fluid. The normal range is 400-1500 ml.
- It is defined based on ultrasound scanning as a pool depth of amniotic fluid of less than 2 cm or amniotic fluid index of less than 5 cm

Causes:

- reduced production of amniotic fluid
    - Failing placental function
    - Bilateral renal agenesis
    - Posterior urethral valves
- Increased losses of amniotic fluid, as in prolonged rupture of membranes

It is common in pregnancies which have progressed many weeks beyond term.

Investigation:

- All cases of oligohydramnios require:
    - Investigation of placental function
    - Ultrasound examination of the fetal kidneys and urogenital system

Complications:

- Complications are as follows:
    - If the oligohydramnios is severe there may be pulmonary hypoplasia
    - The fetus has little room to move; as a result, an early breech presentation is more likely to deliver as a breech.
    - Limb deformities are more common, including talipes and ankylosis of joints
    - Amniotic adhesions may constrict a limb

TOPIC: POLYHYDRAMNIOS

- Polyhydramnios is a common obstetric complication.
- Polyhydramnios is a condition where there is too much amniotic fluid. The normal range is 400-1500 ml with an average of 800 ml.
- On clinical examination, only volumes in excess of 2000 ml are reliably detected.
- Polyhydramnios is defined based on ultrasound scan as the presence of amniotic fluid pool depth of greater than 10 cm or amniotic fluid index more than 25 cm

Causes:

- **Foetal causes:**
    - twin pregnancy, especially uniovular twins
    - anencephaly interferes with fetal swallowing
    - oesophageal or duodenal atresia prevents fetal swallowing
    - spina bifida
    - chorioangioma of the placenta
    - hydrops fetalis
- **Maternal causes:**
    - poorly controlled maternal diabetes results in fetal polyuria
    - multiple pregnancy
- **Idiopathic**: In 65% of cases of polyhydramnios no cause is identified. Idiopathic polyhydramnios is usually mild.

Diagnosis:

The diagnosis of polyhydramnios is largely a clinical one. In cases of polyhydramnios the practitioner may find:

- symptomatology - in acute polyhydramnios the rapid build-up of pressure on surrounding structures may cause shortness of breath, oedema and oliguria
- a tense uterus which is large-for-dates
- difficulty in palpation of fetal parts
- difficulty in hearing fetal heart sounds
- the presence of a fluid thrill
- foetus has an unstable lie

Management:

This is dependent on the onset - acute or chronic - and the magnitude of the polyhydramnios.

- In acute polyhydramnios the mother is usually admitted to hospital and an ultrasound, is taken in order to exclude fetal abnormality. If the fetus appears normal then steps are taken to try and prevent premature labour e.g. the use of prostaglandin antagonists. Diuretics, water and salt restriction seem to be unhelpful and potentially dangerous.

In chronic polyhydramnios an ultrasound investigation and a glucose tolerance test are undertaken. If the polyhydramnios is not severe then the patient is not admitted to hospital.

TOPIC: LEG CRAMPS

Leg cramps — painful involuntary muscle contractions that typically affect the calf, foot or both — are common during pregnancy, often striking at night during the second and third trimesters.

Prevention:

- **Stretch your calf muscles.** Although evidence is lacking, stretching before bed might help prevent leg cramps during pregnancy. Stand at arm's length from a wall, place your hands on the wall in front of you and move your right foot behind your left foot. Slowly bend your left leg forward, keeping your right knee straight and your right heel on the floor. Hold the stretch for about 30 seconds, being careful to keep your back straight and your hips forward. Don't rotate your feet inward or outward. Switch legs and repeat.
- **Stay active.** Regular physical activity might help prevent leg cramps during pregnancy. Before you begin an exercise program, make sure you have your health care provider's OK.
- **Take a magnesium supplement.** Limited research suggests that taking a magnesium supplement might help prevent leg cramps during pregnancy. Make sure you have your health care provider's OK to take a supplement. You might also consider eating more magnesium-rich foods, such as whole grains, beans, dried fruits, nuts and seeds.
- **Stay hydrated.** Keeping your muscles hydrated might help prevent cramps. Your urine should be relatively clear or light yellow in color if you are properly hydrated. If your urine is darker yellow, it might mean that you're not getting enough water.
- **Get adequate calcium.** Some research suggests reduced levels of calcium in your blood during pregnancy may contribute to leg cramps. All women, including pregnant women, should get 1,000 milligrams of calcium a day.
- **Choose proper footwear.** Choose shoes with comfort, support and utility in mind. It might help to wear shoes with a firm heel counter — the part of the shoe that surrounds the heel and helps lock the foot into the shoe.

Stopping the cramp:
If a leg cramp strikes, stretch the calf muscle on the affected side. Walking and then elevating your legs might help keep the leg cramp from returning. A hot shower, warm bath, ice massage or muscle massage also might help.

TOPIC: LOWER-EXTREMITY SWELLING IN LATE PREGNANCY

Edema is common during late pregnancy. It typically involves the lower extremities but occasionally appears as swelling or puffiness in the face or hands.

Causes:

The most common cause of edema in pregnancy is

- Physiologic edema

Physiologic edema results from hormone-induced sodium retention. Edema may also occur when the enlarged uterus intermittently compresses the inferior vena cava during recumbency, obstructing outflow from both femoral veins.

Pathologic causes of edema are less common but often dangerous. **They include:**

- (DVT), preeclampsia, cellulitis, peripartum cardiomyopathy

DVT is more common during pregnancy because pregnancy is a hypercoagulable state, and women may be less mobile.

Preeclampsia results from pregnancy-induced hypertension; however, not all women with preeclampsia develop edema.

Peripartum cardiomyopathy can cause other nonspecific symptoms of pregnancy, including dyspnea and fatigue.

When extensive, cellulitis, which usually causes focal erythema, may resemble general edema.

Evaluation:

Evaluation of patients with lower-extremity edema during late pregnancy aims to exclude DVT, preeclampsia, peripartum cardiomyopathy, and other pathologic causes of edema. Physiologic edema is a diagnosis of exclusion.

History:

History of present illness should include symptom onset and duration, exacerbating and relieving factors (physiologic edema is reduced by lying in the left lateral decubitus position), and risk factors for DVT, preeclampsia, and peripartum cardiomyopathy.

Risk factors for DVT include:

- Venous insufficiency, trauma, cigarette smoking, hypercoagulability disorder
- Thrombotic disorders, immobility, cancer, obesity

Risk factors for preeclampsia include:

- Chronic HTN, first pregnancy, DM, multifetal pregnancy
- Personal or family history of preeclampsia, vascular disorders
- Age < 17 or > 35, obesity, hydatidiform mole, abnormal maternal serum screening results

Risk factors for peripartum cardiomyopathy include:

- Age > 30, hx of cardiomyopathy, African descent, multifetal pregnancy
- History of HTN and/or (eg, preeclampsia)

Review of symptoms:

should seek symptoms of possible causes, including the following:

- Nausea and vomiting, abdominal pain, and jaundice: preeclampsia
- Pain, redness, or warmth in an extremity: DVT or cellulitis
- Dyspnea: Pulmonary edema, peripartum cardiomyopathy, or preeclampsia
- Sudden increase in weight or edema of the hands and face: Preeclampsia
- Headache, confusion, mental status changes, blurry vision, or seizures: Preeclampsia

Past medical history
should include history of DVT, pulmonary embolism, cardiac disease, preeclampsia, and hypertension.

Physical examination:

- Examination begins with review of vital signs, particularly blood pressure.
- Areas of edema are evaluated for distribution (ie, whether bilateral and symmetric or unilateral) and presence of redness, warmth, and tenderness.
- General examination focuses on systems that may show findings of preeclampsia. Eye examination includes testing visual fields for deficits, and funduscopic examination should check for papilledema.
- Cardiovascular examination includes auscultation of the heart and lungs for evidence of fluid overload (eg, audible S3 or S4 heart sounds, tachypnea, rales, crackles) and inspection of neck veins for jugular venous distention. The abdomen should be palpated for tenderness, especially in the epigastric or right upper quadrant region. Neurologic examination should assess mental status for confusion and seek focal neurologic deficits.

Red flags:
The following findings are of particular concern:

- Blood pressure ≥ 140/90 mm Hg
- Unilateral leg or calf warmth, redness, or tenderness, with or without fever
- Systemic symptoms or signs of preeclampsia, particularly mental status changes
- Symptoms or signs of pulmonary edema

Interpretation of findings:
Although edema is common during pregnancy, considering and ruling out the most dangerous causes (preeclampsia, peripartum cardiomyopathy, and DVT) are important:

- If blood pressure is > 140/90 mm Hg, preeclampsia should be considered.
- If edema involves only one leg, particularly when redness, warmth, and tenderness are present, DVT and cellulitis should be considered.
- Bilateral leg edema suggests a physiologic process, preeclampsia, or peripartum cardiomyopathy as the cause.

- Symptoms or signs of pulmonary edema, particularly in patients who have preeclampsia (or other risk factors), suggest peripartum cardiomyopathy.

Testing:

- If preeclampsia is suspected, urine protein is measured; hypertension plus proteinuria indicates preeclampsia. Urine dipstick testing is used routinely, but if diagnosis is unclear, urine protein may be measured in a 24-hour collection. Many laboratories can more rapidly assess urine protein by measuring and calculating the urine protein:urine creatinine ratio. Proteinuria is no longer required to diagnose preeclampsia; abnormal laboratory or clinical findings of preeclampsia in patients with hypertension may also confirm preeclampsia.

- If DVT is suspected, lower-extremity duplex ultrasonography is done.

- If peripartum cardiomyopathy is suspected, ECG, and echocardiography are done. BNP or NT-proBNP is measured.

Treatment:

Specific causes of edema during pregnancy are treated.

Physiologic edema can be reduced by the following:

- Intermittently lying on the left side (left lateral decubitus position), which moves the uterus off the inferior vena cava

- Intermittently elevating the lower extremities

- Wearing elastic compression stockings

- Wearing loose clothing that does not restrict blood flow, particularly in the legs

Diagnosis and treatment of DVT in pregnancy:

- Any woman with symptoms and/or signs suggestive of VTE should have objective testing performed expeditiously and treatment with low-molecular-weight heparin (LMWH) given until the diagnosis is excluded by objective testing, unless treatment is strongly contraindicated
- Diagnosis of an acute DVT:
    - Compression duplex ultrasound should be undertaken where there is clinical suspicion of DVT
    - If ultrasound is negative and there is a low level of clinical suspicion, anticoagulant treatment can be discontinued
    - If ultrasound is negative and a high level of clinical suspicion exists, anticoagulant treatment should be discontinued but the ultrasound should be repeated on days 3 and 7

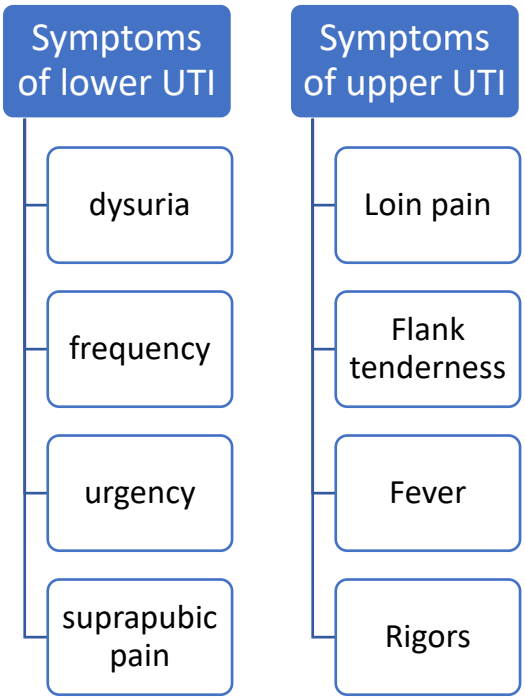
Key points:

- Edema is common and usually benign (physiologic) during late pregnancy.

- Physiologic edema is reduced by lying in the left lateral decubitus position, elevating the lower extremities, using compression stockings, and wearing loose clothing that does not restrict blood flow.

TOPIC: Urinary tract infection

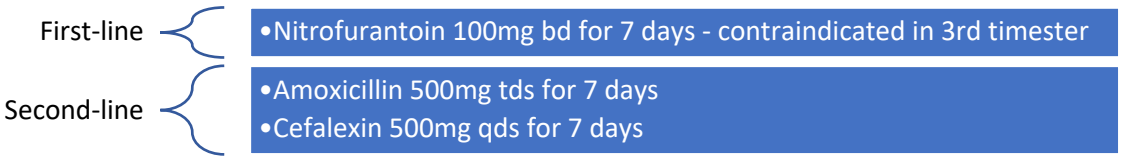
Features:



Investigations:

- All pregnant women with suspected UTI should have urine culture sent and started on antibiotics

Management:



Referral:

- Failure of treatment
- Frank hematuria
- Recurrent UTI's